SHRINK THE DUAL REGULARIZATION, KEEP THE PENALTY: TIGHT COMPLEXITY OF LINEARIZED AUGMENTED LAGRANGIAN METHODS

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Abstract. We study single-loop linearized augmented Lagrangian methods for the nonconvex composite problem $\min_{\boldsymbol{x}} f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x}))$, where f and $\mathcal{A}$ are smooth and h is convex and Lipschitz. We consider a family of methods that perform one gradient descent–ascent step per iteration on the reduced augmented Lagrangian with a fixed penalty ρ, a dual regularization with parameter s, and moving primal and dual centers. Every member of this family has the effective penalty $\rho + s$. We show that the iteration complexity of the family for finding an ϵ-KKT point is determined by $\rho + s$ alone; the center stepsizes do not change its order. If $\epsilon(\rho + s)$ is at most a constant, then for any arbitrary choice of constant center stepsizes there is an instance on which the method, started from zero, requires $\Omega(\epsilon^{-4})$ iterations, and this order is attained within the family. If ρ is fixed and s is of order $1/\epsilon$, a member of the family finds an ϵ-KKT point in $\mathcal{O}(\epsilon^{-3})$ iterations. Hence the effective penalty must reach the order $1/\epsilon$ to improve on ϵ^{-4}, and this growth can be carried by the dual regularization alone, with the penalty kept fixed.

Key words. nonsmooth composite optimization, augmented Lagrangian method, dual-regularization, iteration complexity

AMS subject classifications. 65K05, 90C26, 90C30, 90C46

1. Introduction.

In this paper, we study the structured nonsmooth nonconvex composite problem

$$\text{(P)} \qquad \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})),$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a possibly nonconvex function with a Lipschitz continuous gradient, $\mathcal{A}\colon \mathbb{R}^n \to \mathbb{R}^m$ is a smooth and possibly nonlinear mapping, and $h\colon \mathbb{R}^m \to \mathbb{R}\cup\{+\infty\}$ is a proper closed convex function whose proximal mapping is easy to evaluate; the complete assumptions are stated in Section 2.1. The two components play complementary roles: f is usually a smooth loss or data-fidelity term, whereas h is a nonsmooth convex function that imposes robustness, sparsity, or other structure on the transformed variable $\mathcal{A}(\boldsymbol{x})$ [4, 23, 26]. Several models of practical interest take this form.

1.1. Motivating Examples.

Robust phase retrieval [6]. Given measurement vectors $\mathbf{a}_i \in \mathbb{R}^n$ and observations $b_i \approx \langle \mathbf{a}_i, \boldsymbol{x}^\star\rangle^2$, a fraction of which may be grossly corrupted, one solves

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \ \frac{1}{m}\sum_{i=1}^{m} \big|\langle \mathbf{a}_i, \boldsymbol{x}\rangle^2 - b_i\big|.$$

This is Problem (P) with $f \equiv 0$, $h = \frac{1}{m}\|\cdot\|_1$, and $\mathcal{A}(\boldsymbol{x})_i = \langle \mathbf{a}_i, \boldsymbol{x}\rangle^2 - b_i$; the ℓ_1 loss makes the model robust to outliers.

Robust low-rank matrix recovery [4]. Given a linear measurement map $\mathcal{M}\colon \mathbb{R}^{d_1\times d_2} \to \mathbb{R}^m$ and observations $\mathbf{b}$ containing outliers, a rank-r matrix is recovered in factorized form via

$$\min_{\mathbf{U}\in\mathbb{R}^{d_1\times r},\, \mathbf{V}\in\mathbb{R}^{d_2\times r}} \ \frac{1}{m}\big\|\mathcal{M}(\mathbf{U}\mathbf{V}^\top) - \mathbf{b}\big\|_1.$$

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Here $\boldsymbol{x} = (\mathbf{U}, \mathbf{V})$, $h = \frac{1}{m}\|\cdot\|_1$, and $\mathcal{A}(\boldsymbol{x}) = \mathcal{M}(\mathbf{U}\mathbf{V}^\top) - \mathbf{b}$ is bilinear in $(\mathbf{U}, \mathbf{V})$, so the model again has the form Problem (**P**).

Minimizing a pointwise maximum [36]. For smooth losses $f_1, \ldots, f_m$, such as the losses on m data subgroups, one solves

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \max_{1\le i\le m} f_i(\boldsymbol{x}).$$

This is Problem (**P**) with $h(u) = \max_{1\le i\le m} u_i$ and $\mathcal{A}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x}))$; a smooth regularizer, if present, enters through f, and the proximal mapping of h reduces to a projection onto the unit simplex.

1.2. Related Work. Introducing the auxiliary variable $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$ rewrites Problem (**P**) as the equality-constrained problem

$$\min_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{x}) + h(\boldsymbol{y}) \quad \text{s.t.} \quad \mathcal{A}(\boldsymbol{x}) = \boldsymbol{y}, \tag{1.1}$$

which brings it within the scope of penalty methods and augmented Lagrangian methods (ALMs). For general nonlinearly constrained nonconvex problems, both families control feasibility through the penalty parameter. Quadratic penalty methods increase the penalty without bound [13, 17]. ALMs were originally introduced to avoid such unbounded penalties [11, 27], and in convex programming any fixed penalty suffices [29]; in the nonconvex setting, however, inexact ALMs still increase the penalty across outer iterations to obtain complexity guarantees [31, 16], and proximal ALM, which keeps the penalty fixed within a run, chooses its value according to the target accuracy [33]. Single-loop methods have attracted growing attention because they avoid solving subproblems to high accuracy, yet they rely on the same mechanism: single-loop quadratic penalty schemes increase the penalty [1], and linearized ALMs use small dual steps oraccuracy *eps*-dependent parameters [32]. A large penalty inflates the smoothness constant of the primal update, which forces small step sizes and makes the primal subproblems increasingly ill-conditioned.

The split formulation (1.1) also carries favorable structure. Its constraint Jacobian $[J_{\mathcal{A}}(\boldsymbol{x}), -I]$ has full row rank at every $\boldsymbol{x}$, so constraint qualifications hold automatically, and its objective separates into a smooth part in $\boldsymbol{x}$ and a prox-friendly part in $\boldsymbol{y}$. Nonconvex ADMM and related Lagrangian-based splitting methods exploit exactly this structure [2, 5, 9, 8]. The structure, however, does not remove the need for a large penalty. For nonlinear $\mathcal{A}$, adaptive Lagrangian schemes require penalty levels proportional to $1/\epsilon$ [9], and linearized perturbed ALMs use accuracy-dependent penalty and perturbation parameters [8]. The nonconvex–linear minimax route attains $\mathcal{O}(\epsilon^{-3})$ [25, 10, 34], but its dual regularization acts as an effective penalty that grows as the target accuracy decreases. Fixed-penalty guarantees are available for smooth nonlinear equality constraints under a uniform LICQ [7, 19], but they do not cover the nonsmooth term $h(\boldsymbol{y})$ in (1.1). It therefore remains open whether a single-loop ALM can keep every penalty-type parameter independent of both the accuracy and the target accuracy while retaining an explicit first-order update for $h \circ \mathcal{A}$.

From another perspective, Problem (**P**) also admits a minimax reformulation. In particular, eliminating $\boldsymbol{y}$ from the Lagrangian of (1.1) gives

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \max_{\boldsymbol{\lambda}\in\mathbb{R}^m} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \Phi(\boldsymbol{x}, \boldsymbol{\lambda}) - h^*(\boldsymbol{\lambda}), \qquad \Phi(\boldsymbol{x}, \boldsymbol{\lambda}) := f(\boldsymbol{x}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x})\rangle, \tag{1.2}$$

where $\operatorname{dom} h^* \subseteq \mathbb{B}(\mathbf{0}, L_h)$ by Assumption 2.2. This is a nonconvex–linear instance of the smooth nonconvex–concave (NC-C) minimax problem. Since $\boldsymbol{\lambda}$ is exactly the multiplier of $\mathcal{A}(\boldsymbol{x}) = \boldsymbol{y}$, KKT points of (1.1) and game-stationary (GS) points of (1.2) agree up to constants, and ALMs can be read

as single-loop minimax methods. For large-scale problems, we focus on single-loop first-order methods, which update each variable by one explicit step per iteration without an inner solver. The basic method of this kind for (1.2) is gradient descent–ascent (GDA). It converges slowly because regularity fails on both sides: Φ is only linear in $\boldsymbol{\lambda}$, so the best response $\boldsymbol{\lambda}^\star(\boldsymbol{x})$ is in general set-valued, and Φ is nonconvex in $\boldsymbol{x}$, so $\boldsymbol{x}^\star(\boldsymbol{\lambda})$ need not be unique or stable. Even with two timescales, GDA needs $\mathcal{O}(\epsilon^{-6})$ iterations to reach an ϵ-GS point [18].

Two techniques improve on this rate, each restoring regularity on one side. The first is primal smoothing, which adds $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$ with a fixed γ and moves the center slowly by $\boldsymbol{z}^{t+1} = \boldsymbol{z}^t+\beta(\boldsymbol{x}^{t+1}-\boldsymbol{z}^t)$. The target accuracy enters only through β: with $\beta = \Theta(\epsilon^2)$, the resulting Smoothed GDA finds an ϵ-GS point in $\mathcal{O}(\epsilon^{-4})$ iterations. For the pointwise maximum of finitely many smooth functions, i.e., (1.2) with $h(\boldsymbol{u}) = \max_i u_i$, strict complementarity permits a constant β and improves the bound to $\mathcal{O}(\epsilon^{-2})$ [36].

The second technique is dual regularization, which subtracts $\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ from the dual objective and makes it strongly concave. In our reduced augmented-Lagrangian formulation, this regularization has an effective-penalty interpretation:

$$\max_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) = \mathcal{D}_{\rho+s}(\boldsymbol{x},\boldsymbol{\mu}) + \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2, \tag{1.3}$$

where $\mathcal{P}_s$ is defined in (3.3) and $\mathcal{D}_{\rho+s}$ is the reduced augmented Lagrangian (3.1) with penalty $\rho+s$; see Lemma 3.1. A zero center $\boldsymbol{\mu} = \mathbf{0}$ recovers a quadratic penalty with coefficient $\rho+s$, which can leave a nonzero feasibility residual when s is fixed. Updating $\boldsymbol{\mu}$ instead provides a multiplier-update mechanism that may recover feasibility while keeping $\rho+s$ fixed; its convergence and complexity depend on the coupled updates.

With the dual center fixed at $\boldsymbol{\mu} = \mathbf{0}$, Perturbed GDA attains an $\mathcal{O}(\epsilon^{-4})$ complexity for finding an ϵ-GS point [21, 35], and Li et al. establish a matching algorithm-specific lower bound [14]. Methods exploiting the NC-L structure attain $\mathcal{O}(\epsilon^{-3})$ [25, 10, 34]. These guarantees rely on accuracy-dependent regularization parameters, corresponding to an effective penalty that increases as ϵ decreases. To improve the iteration complexity, Li et al. combine primal smoothing with zero-centered dual perturbation to develop Perturbed Smoothed GDA, which yields a genuine improvement in complexity order, reducing the GS complexity of Perturbed GDA from $\mathcal{O}(\epsilon^{-4})$ to $\mathcal{O}(\epsilon^{-3})$ with a matching lower bound [14]. However, the dual regularization coefficient still satisfies $1/s = \Theta(\epsilon)$.

With a moving dual center $\boldsymbol{\mu}$ and fixed primal and dual regularization coefficients, DS-GDA achieves $\mathcal{O}(\epsilon^{-4})$ complexity for nonconvex–concave problems. Faster rates are available under a uniform one-sided KŁ condition [39]. Without primal smoothing, PPALA reports $\widetilde{\mathcal{O}}(\epsilon^{-2})$ complexity with fixed parameters for functionally constrained optimization [12]. However, its feasibility argument relies on asymptotic reasoning, and a corresponding finite-iteration feasibility bound remains to be verified.

In all guarantees above that reach $\mathcal{O}(\epsilon^{-3})$, some parameter grows with $1/\epsilon$: either the penalty, as in penalty methods and GDPA, or the dual regularization, as in Perturbed Smoothed GDA. Practical ALM implementations, in contrast, usually keep the penalty fixed. Growing the penalty is costly for a single-loop method, because the Lipschitz constant of $\nabla_{\boldsymbol{x}}\mathcal{D}_\rho$ increases with ρ and forces the primal stepsize to shrink. This leads to the following question.

> **(Q)** *To reach $\mathcal{O}(\epsilon^{-3})$, must a single-loop linearized ALM let some parameter grow with $1/\epsilon$? If so, must this growth be carried by the penalty parameter?*

We answer the first part affirmatively and the second part negatively. By Lemma 3.1, the relevant quantity is the effective penalty $\rho+s$. For any arbitrary choice of constant primal and dual center stepsizes, SPLALM requires $\Omega(\epsilon^{-4})$ iterations whenever $\rho+s \le c_1/\epsilon$ (Theorem 4.2). Conversely, with ρ fixed, s of order $1/\epsilon$, and the dual center fixed at zero, SPLALM reduces to perturbed smoothed

Algorithms	Regularization and penalty parameters				Complexity	
	Penalty parameter	Primal smoothing $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$	Dual regularization $-\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$		Upper bound	Lower bound
			Center $\boldsymbol{\mu}$	Parameter s		
GDA-based Methods						
Smoothed GDA [37, 14]	—	$\gamma=\Theta(1)$	—	—	$\mathcal{O}(\epsilon^{-4})$	$\Omega(\epsilon^{-4})$
Perturbed GDA [21, 35, 14]	—	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-4})$	$\Omega(\epsilon^{-4})$
Perturbed Smoothed GDA [14]	—	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-3})$	$\Omega(\epsilon^{-3})$
RADA [34]	—	—	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$(r_\epsilon+\beta_k)^{-1}$	$\mathcal{O}(\epsilon^{-3})$	—
DS-GDA [39]	—	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta(1)$	$\underline{\mathcal{O}(\epsilon^{-4})}$	—
Penalty and augmented Lagrangian-based methods						
Quadratic penalty [1]	$\rho_k=\Theta((k+1)^{1/4})$	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta((k+1)^{1/4})$	$\widetilde{\mathcal{O}}(\epsilon^{-4})$	—
GDPA [20]	$\rho_k=\Theta(\epsilon^{-1})$	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-3})$	—
PPALA [12]	$\rho=\Theta(1)$	—	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta(1)$	$\widetilde{\mathcal{O}}(\epsilon^{-2})$	—
SPLALM *Theorem 4.2*	$\rho>0$	$\gamma=\Theta(1)$	any	$\rho+s\le c_1\epsilon^{-1}$	?	$\Omega(\epsilon^{-4})$
SPLALM *Theorem C.1*	$\rho=\Theta(1)$	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta((k+1)^{1/3})$	$\mathcal{O}(\epsilon^{-3})$	?

Table 1: Representative single-loop methods, grouped by their original formulation. The constrained rows concern nonlinear constraints. Upper bounds display the dependence on ϵ, suppressing problem constants and bounded initial gaps. The regularization columns specify parameters and centers, not gradient stepsizes; primal smoothing uses a separate moving center. "—" denotes an absent regularizer or an unreported lower bound; question marks denote results to be established. Constant parameters are independent of ϵ and k. The rates use each method's stated stationarity criterion and assumptions; lower bounds are algorithm-specific.

GDA applied to $\mathcal{D}_\rho$ and reaches an ϵ-KKT point in $\mathcal{O}(\epsilon^{-3})$ iterations [14, Theorem 3.3]. The growth required by the lower bound can thus be carried entirely by the dual regularization, while the penalty stays fixed. Appendix C gives a schedule with increasing s_k that attains the same rate without fixing ϵ in advance. Table 1 summarizes representative complexity guarantees.

In this paper, we provide a precise characterization addressing (Q). Our contributions can be summarized as follows.

1. **A lower bound independent of the center stepsizes.** We study SPLALM, which performs alternating gradient descent–ascent steps on the reduced augmented Lagrangian $\mathcal{D}_\rho$ with a fixed penalty $\rho > 0$, a dual regularization with parameter s, and moving primal and dual centers (Definition 3.1). The family contains the update structures of quadratic-penalty, perturbed smoothed GDA, doubly smoothed GDA, smoothed GDA, and linearized ALM methods acting on $\mathcal{D}_\rho$ (Table 2), and every member has the effective penalty $\rho + s$ (Lemma 3.1). By carefully designing hard instances, we establish a lower bound of $\Omega(\epsilon^{-4})$ for SPLALM with any arbitrary choice of constant center stepsizes $(\alpha, \beta) \in [0,1]^2$, whenever $\epsilon(\rho + s) \le c_1$ (Theorem 4.2). On each hard instance, SPLALM started from zero requires at least this many iterations to guarantee that one of its outputs is an ϵ-KKT point. Hence moving the centers cannot compensate for an effective penalty $\rho + s = o(\epsilon^{-1})$.
2. **The necessary scale of the effective penalty.** We demonstrate that, with ρ fixed, $s = \Theta(\epsilon^{-1})$, and $\beta = 0$, SPLALM reduces to perturbed smoothed GDA applied to $\mathcal{D}_\rho$ and finds an ϵ-KKT point within $\mathcal{O}(\epsilon^{-3})$ iterations. The result follows from [14, Theorem 3.3] once the initial gap and the stationarity measure are expressed through the constants of $\mathcal{I}(\mathbf{C}, \Delta)$. Together with the lower bound, this establishes ϵ^{-1} as the order that the effective penalty must reach, and suffices to reach, for constant parameters. In other words, the dual regularization alone can carry this growth while the penalty stays fixed.
3. **Tightness under a fixed effective penalty.** For $\beta = 1$, SPLALM reduces to smoothed GDA applied to $\mathcal{D}_\rho$, and we establish the iteration complexity $\mathcal{O}(\epsilon^{-4})$ uniformly over $\mathcal{I}(\mathbf{C}, \Delta)$ by adapting [37]. This certifies that the lower bound is tight for fixed ρ and s and sufficiently small γ, η, and τ. These results lead to the following conceptual picture: the order of the iteration complexity is determined by $\rho + s$ alone, and the center stepsizes do not change it.

Organization. Section 2 introduces the problem class and the KKT and game-stationarity criteria. Section 3 introduces the SPLALM family and the complexity measure, and shows that every member of the family has the effective penalty $\rho + s$. Section 4 states the main results, and Section 5 proves the lower bound.

2. Preliminaries. Notation. We use $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ for the Euclidean inner product and norm; for a matrix, $\| \cdot \|$ is the spectral norm. For $\mathcal{A} = (A_1, \dots, A_m)^\top$, the Jacobian $J_{\mathcal{A}}(\boldsymbol{x}) \in \mathbb{R}^{m \times n}$ has ith row $\nabla A_i(\boldsymbol{x})^\top$. For a nonempty closed convex set C, $\operatorname{dist}(\boldsymbol{u}, C) := \min_{\boldsymbol{v} \in C} \|\boldsymbol{u} - \boldsymbol{v}\|$, and $\Pi_C(\boldsymbol{u})$ is the Euclidean projection of $\boldsymbol{u}$ onto C. When convenient, we identify $\mathbb{R}^2$ with $\mathbb{C}$, so that $\|p\| = |p|$ and $\langle p, q \rangle = \operatorname{Re}(\bar{p} q)$.

For a proper closed convex function $h : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$, its subdifferential, convex conjugate, and proximal mapping are

$$\partial h(\boldsymbol{y}) := \{\boldsymbol{v} : \ h(\boldsymbol{w}) \ge h(\boldsymbol{y}) + \langle \boldsymbol{v}, \boldsymbol{w} - \boldsymbol{y} \rangle \text{ for all } \boldsymbol{w}\}, \qquad h^*(\boldsymbol{q}) := \sup_{\boldsymbol{y} \in \mathbb{R}^m} \{\langle \boldsymbol{q}, \boldsymbol{y} \rangle - h(\boldsymbol{y})\},$$

$$\operatorname{prox}_{th}(\boldsymbol{u}) := \operatorname*{argmin}_{\boldsymbol{y} \in \mathbb{R}^m} \Big\{ h(\boldsymbol{y}) + \frac{1}{2t} \|\boldsymbol{y} - \boldsymbol{u}\|^2 \Big\}, \qquad t > 0; \tag{2.1}$$

see [28]. We write $\operatorname{dom} \partial h^* := \{\boldsymbol{q} : \partial h^*(\boldsymbol{q}) \neq \varnothing\}$.

2.1. Assumptions. We impose the following assumptions.

Assumption 2.1 (Smooth components). There exist constants $L_f, G_f, L_{\mathcal{A}}, G_{\mathcal{A}} \geq 0$ such that, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^n$,

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{x}')\| \leq L_f \|\boldsymbol{x} - \boldsymbol{x}'\|, \quad \|\nabla f(\boldsymbol{x})\| \leq G_f,$$
$$\|J_{\mathcal{A}}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')\| \leq L_{\mathcal{A}} \|\boldsymbol{x} - \boldsymbol{x}'\|, \quad \|J_{\mathcal{A}}(\boldsymbol{x})\| \leq G_{\mathcal{A}}.$$

Assumption 2.2 (Lipschitz outer function). The function $h : \mathbb{R}^m \to \mathbb{R}$ is convex and L_h-Lipschitz:

$$|h(\boldsymbol{u}) - h(\boldsymbol{v})| \leq L_h \|\boldsymbol{u} - \boldsymbol{v}\|, \qquad \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m.$$

2.2. Problem class, stationarity, and oracle. To study (Q), we introduce the following problem class.

DEFINITION 2.1 (Problem class). *Let $\Delta > 0$ and $\mathbf{C} = (L_f, G_f, L_{\mathcal{A}}, G_{\mathcal{A}}, L_h) \in \mathbb{R}^5_{\geq 0}$. We denote by $\mathcal{I}(\mathbf{C}, \Delta)$ the set of all triples $(f, \mathcal{A}, h)$, over all dimensions $n, m \in \mathbb{N}$, such that Assumptions 2.1 and 2.2 hold with the constants in $\mathbf{C}$ and $F(\mathbf{0}) - \inf_{\boldsymbol{x} \in \mathbb{R}^n} F(\boldsymbol{x}) \leq \Delta$.*

The constants $\mathbf{C}$ and Δ do not depend on the target accuracy or on n, m, and the gap condition implies $\inf F > -\infty$.

We use the following stationarity measure for Problem (**P**).

DEFINITION 2.2 (ϵ-KKT point). *Given $\epsilon > 0$. A triple $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$ with $\boldsymbol{q} \in \partial h(\boldsymbol{y})$ is an ϵ-KKT point of (**P**) if*

$$\mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q}) := \max\Big\{ \big\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}\big\|,\ \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| \Big\} \leq \epsilon \tag{2.2}$$

We next relate Definition 2.2 to game stationarity (GS) for the minimax reformulation Problem (1.2) of Problem (**P**). We use the following GS definition from [15, Definition 3.1(b)] and [38, (2.10)–(2.11)].

DEFINITION 2.3 (ϵ-GS point). *Let $\epsilon > 0$. A pair $(\boldsymbol{x}, \boldsymbol{q}) \in \mathbb{R}^n \times \mathrm{dom}\partial h^*$ is an ϵ-game stationary point (ϵ-GS) of Problem (1.2) if*

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{q}) := \max\Big\{ \big\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}\big\|,\ \mathrm{dist}\big(\mathcal{A}(\boldsymbol{x}), \partial h^*(\boldsymbol{q})\big) \Big\} \leq \epsilon.$$

We now relate ϵ-KKT points of Problem (**P**) to ϵ-GS points of its Fenchel saddle reformulation Problem (1.2).

PROPOSITION 2.1 (KKT points of Problem (**P**) and GS points of Problem (1.2)). *Suppose that Assumption 2.2 holds, and let $(\boldsymbol{x}, \boldsymbol{q}) \in \mathbb{R}^n \times \mathrm{dom}\partial h^*$. Then the following measurement relationship holds:*

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{q}) = \min_{\boldsymbol{y} \in \partial h^*(\boldsymbol{q})} \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q}),$$

with minimum attained at $\hat{\boldsymbol{y}} = \Pi_{\partial h^*(\boldsymbol{q})}\big(\mathcal{A}(\boldsymbol{x})\big)$. Consequently, for every $\epsilon > 0$:

*(a) if $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$ is an ϵ-KKT point of Problem (**P**), then $(\boldsymbol{x}, \boldsymbol{q})$ is an ϵ-GS point of Problem (1.2);*

*(b) if $(\boldsymbol{x}, \boldsymbol{q})$ is an ϵ-GS point of Problem (1.2), then $(\boldsymbol{x}, \hat{\boldsymbol{y}}, \boldsymbol{q})$ is an ϵ-KKT point of Problem (**P**).*

Proof. Since h is proper, closed, and convex, $\boldsymbol{q} \in \partial h(\boldsymbol{y})$ if and only if $\boldsymbol{y} \in \partial h^*(\boldsymbol{q})$ [28, Theorem 23.5]. The first term of $\mathcal{R}_{\mathrm{KKT}}$ does not depend on $\boldsymbol{y}$. The set $\partial h^*(\boldsymbol{q})$ is nonempty, closed, and convex, so $\min_{\boldsymbol{y} \in \partial h^*(\boldsymbol{q})} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| = \mathrm{dist}(\mathcal{A}(\boldsymbol{x}), \partial h^*(\boldsymbol{q}))$, attained at $\hat{\boldsymbol{y}}$. Statements (a) and (b) follow. ∎

We next introduce the following deterministic first-order oracle model.

DEFINITION 2.4 (Oracle model). *We assume access to a deterministic first-order oracle*

$$\mathcal{O}_{f,\mathcal{A}}(\boldsymbol{x}) := \big(f(\boldsymbol{x}), \nabla f(\boldsymbol{x}), \mathcal{A}(\boldsymbol{x}), J_{\mathcal{A}}(\boldsymbol{x})\big),$$

and an exact proximal oracle

$$\mathcal{O}_h(\boldsymbol{v}, t) := \operatorname{prox}_{th}(\boldsymbol{v}), \qquad t > 0.$$

Each call to either oracle counts as one oracle evaluation. In particular, the cost of solving the proximal subproblem is represented by one exact proximal-oracle call.

3. Algorithm Family and Effective Penalty.

3.1. Algorithm family and worst-case iteration complexity. For $\rho > 0$, the reduced augmented Lagrangian of (1.1) is

$$\mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) := \min_{\boldsymbol{y} \in \mathbb{R}^m} \Big\{ f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho}{2} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2 \Big\}. \tag{3.1}$$

Its minimizer and the associated multiplier are

$$\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}) := \operatorname{prox}_{h/\rho}\Big(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\Big), \qquad \boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) := \boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda})\big], \tag{3.2}$$

and $\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) \in \partial h(\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}))$. We study the smoothed perturbed linearized augmented Lagrangian method (SPLALM), which performs alternating gradient descent–ascent steps on the smoothed perturbed potential function

$$\mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{1}{2s}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2. \tag{3.3}$$

The proximal term smooths the primal problem, the term $-\frac{1}{2s}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2$ perturbs the dual problem, and $\boldsymbol{z} \in \mathbb{R}^n$ and $\boldsymbol{\mu} \in \mathbb{R}^m$ are the primal and dual centers.

DEFINITION 3.1 (SPLALM). *Let $\theta = (\rho, s, \gamma, \eta, \tau)$ satisfy $\rho > 0$, $0 < \eta \le \gamma < \infty$, and $0 < \tau \le s < \infty$, and let $(\alpha, \beta) \in [0,1]^2$. The method* $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ *starts from $(\boldsymbol{x}^0, \boldsymbol{\lambda}^0, \boldsymbol{z}^0, \boldsymbol{\mu}^0) = \mathbf{0}$ and, for $k \ge 0$, sets*

$$\begin{aligned}
\boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta\, \nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k),\\
\boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \tau\, \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k),\\
\boldsymbol{z}^{k+1} &= \boldsymbol{z}^k + \alpha(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k),\\
\boldsymbol{\mu}^{k+1} &= \boldsymbol{\mu}^k + \beta(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k).
\end{aligned} \tag{3.4}$$

At iteration k, it outputs $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k) := (\boldsymbol{x}^{k+1}, \boldsymbol{y}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k), \boldsymbol{q}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k))$.

With first-order information cached between iterations, K iterations and their outputs use $K+1$ calls to $\mathcal{O}_{f,\mathcal{A}}$ and $2K$ calls to $\mathcal{O}_h$. Thus the total oracle count is $3K+1$, and iteration and oracle complexity agree up to a factor of three, apart from one initial first-order oracle call.

DEFINITION 3.2 (SPLALM family). *For θ as in Definition 3.1, $\mathcal{C}_\theta := \{\mathrm{SPLALM}_{\theta,\alpha,\beta} : (\alpha, \beta) \in [0,1]^2\}$.*

Remark 3.1 (Initialization). Every method in $\mathcal{C}_\theta$ starts from zero on every instance. Theorem 4.2 constructs a hard instance for each $(\theta, \alpha, \beta, \epsilon)$ while keeping this initialization fixed. Thus, the lower bound holds for the prescribed starting point. The constructions in [14, Section 5.1.1] and [22, Appendix C.1] choose both the hard function and the initialization.

DEFINITION 3.3 (Worst-case iteration complexity). *Let $\epsilon > 0$ and $I \in \mathcal{I}(\mathbf{C}, \Delta)$, and let $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ be the outputs of* $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ *on I. Define*

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) := \inf\Big\{K \geq 1 : \min_{0 \leq k < K} \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k) \leq \epsilon\Big\},$$
$$T_\epsilon^{\mathrm{GS}}(\theta, \alpha, \beta; I) := \inf\Big\{K \geq 1 : \min_{0 \leq k < K} \mathcal{G}(\boldsymbol{x}^{k+1}, \boldsymbol{q}^k) \leq \epsilon\Big\},$$

with $\inf \varnothing = +\infty$. *For $\bullet \in \{\mathrm{KKT}, \mathrm{GS}\}$, the worst-case iteration complexity of $\mathcal{C}_\theta$ over $\mathcal{I}(\mathbf{C}, \Delta)$ is*

$$T_\epsilon^\bullet\big(\mathcal{C}_\theta, \mathcal{I}(\mathbf{C}, \Delta)\big) := \inf_{(\alpha,\beta)\in[0,1]^2} \sup_{I \in \mathcal{I}(\mathbf{C},\Delta)} T_\epsilon^\bullet(\theta, \alpha, \beta; I).$$

The infimum precedes the supremum, so (α, β) may depend on ϵ, θ, and $(\mathbf{C}, \Delta)$, but not on the instance. At least one of the first K outputs meets the tolerance if and only if $K \geq T_\epsilon^\bullet$; hence the definition covers every rule that returns one of these outputs, including the best-so-far rule. By Proposition 2.1, $\mathcal{G}(\boldsymbol{x}^{k+1}, \boldsymbol{q}^k) \leq \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ for every k, so $T_\epsilon^{\mathrm{GS}}(\mathcal{C}_\theta, \mathcal{I}) \leq T_\epsilon^{\mathrm{KKT}}(\mathcal{C}_\theta, \mathcal{I})$. An upper bound for the KKT measure therefore yields the same upper bound for the GS measure, and a lower bound for the GS measure yields the same lower bound for the KKT measure.

Remark 3.2 (Family complexity and oracle complexity). Let $\mathcal{A}_{\mathrm{det}}$ be the class of deterministic methods whose iterates are arbitrary maps of all previous responses of $\mathcal{O}_{f,\mathcal{A}}$ and $\mathcal{O}_h$, defined as in [3, Section 2.2] and [24, Definition 2.5] with the oracles of Definition 2.4. Then $\mathcal{C}_\theta \subset \mathcal{A}_{\mathrm{det}}$, and

$$T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I}) = \inf_{A \in \mathcal{A}_{\mathrm{det}}} \sup_{I \in \mathcal{I}} T_\epsilon(A; I) \leq \inf_{(\alpha,\beta)\in[0,1]^2} \sup_{I \in \mathcal{I}} T_\epsilon(\theta, \alpha, \beta; I) = T_\epsilon(\mathcal{C}_\theta, \mathcal{I}).$$

(i) (Scope) A lower bound on $T_\epsilon(\mathcal{C}_\theta, \mathcal{I})$ does not bound $T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I})$; methods outside $\mathcal{C}_\theta$ may be faster. When h is the support function of a compact convex set, (1.2) is a smooth nonconvex–concave problem, and the $O(\epsilon^{-5/2})$ GS bound of [38] and the $O(\epsilon^{-3})$ OS bound of [24] are attained by methods outside $\mathcal{C}_\theta$.

(ii) (Uniformity) The infimum ranges over all $(\alpha, \beta) \in [0, 1]^2$, including the endpoint cases in Table 2, and Theorem 4.2 holds for all (ρ, s) with $\epsilon(\rho + s) \leq c_1$. A lower bound on $T_\epsilon(\mathcal{C}_\theta, \mathcal{I})$ therefore cannot be removed by tuning the center stepsizes; it is governed by the effective penalty $\rho + s$ of Lemma 3.1.

(iii) (Transfer) Conversely, every lower bound on $T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I})$ for the same problem class and measure also holds for $\mathcal{C}_\theta$.

3.2. Effective penalty and special cases. The reduced augmented Lagrangian is a dual proximal smoothing of the saddle function $\mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) = f(\boldsymbol{x}) + \langle \mathcal{A}(\boldsymbol{x}), \boldsymbol{q}\rangle - h^*(\boldsymbol{q})$ in (1.2) [30]. The center of this smoothing is the multiplier, and its strength is the penalty. Adding the dual regularizer of (3.3) keeps this form; it only raises the penalty and moves the center.

LEMMA 3.1 (Effective penalty). *Suppose that Assumption 2.2 holds. For every $\sigma > 0$,*

$$\mathcal{D}_\sigma(\boldsymbol{x}, \boldsymbol{\lambda}) = \max_{\boldsymbol{q} \in \mathbb{R}^m} \Big\{\mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) - \frac{1}{2\sigma}\|\boldsymbol{q} - \boldsymbol{\lambda}\|^2\Big\}, \tag{3.5}$$

Table 2: Special cases of $\mathcal{C}_\theta$. All methods act on $\mathcal{D}_\rho$ with $\rho > 0$ and have the effective penalty $\rho + s$.

β	Dual center	$\alpha = 1$	$0 < \alpha < 1$
0	$\mathbf{0}$	perturbed GDA [21, 35]	perturbed smoothed GDA [14]
$(0, 1)$	$\sum_{j=1}^{k} \beta(1-\beta)^{k-j}\boldsymbol{\lambda}^j$	—	DS-GDA [39], $r_1 = 1/\gamma$, $r_2 = 1/s$
1	$\boldsymbol{\lambda}^k$	linearized ALM ($\tau = \rho$)	smoothed GDA [37]

and consequently

$$\max_{\boldsymbol{\lambda}\in\mathbb{R}^m} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) = \mathcal{D}_{\rho+s}(\boldsymbol{x}, \boldsymbol{\mu}) + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{z}\|^2. \tag{3.6}$$

Proof. Write $\boldsymbol{u} = \mathcal{A}(\boldsymbol{x})$ and $h(\boldsymbol{y}) = \max_{\boldsymbol{q}}\{\langle \boldsymbol{q}, \boldsymbol{y}\rangle - h^*(\boldsymbol{q})\}$. Since $\mathrm{dom} h^*$ lies in a ball by Assumption 2.2, Sion's minimax theorem allows the minimum over $\boldsymbol{y}$ and the maximum over $\boldsymbol{q}$ in $\min_{\boldsymbol{y}} \max_{\boldsymbol{q}}\{\langle \boldsymbol{q}-\boldsymbol{\lambda}, \boldsymbol{y}\rangle - h^*(\boldsymbol{q}) + \frac{\sigma}{2}\|\boldsymbol{u}-\boldsymbol{y}\|^2\}$ to be interchanged. The inner minimizer is $\boldsymbol{y} = \boldsymbol{u} - (\boldsymbol{q}-\boldsymbol{\lambda})/\sigma$, which gives (3.5). Applying (3.5) with $\sigma = \rho$ and exchanging the two maximizations,

$$\max_{\boldsymbol{\lambda}} \mathcal{P}_s = \max_{\boldsymbol{q}}\Big\{\mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) - \min_{\boldsymbol{\lambda}}\Big[\frac{\|\boldsymbol{q}-\boldsymbol{\lambda}\|^2}{2\rho} + \frac{\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2}{2s}\Big]\Big\} + \frac{\|\boldsymbol{x}-\boldsymbol{z}\|^2}{2\gamma},$$

and the inner minimum equals $\|\boldsymbol{q}-\boldsymbol{\mu}\|^2/(2(\rho+s))$. Then (3.5) with $\sigma = \rho + s$ gives (3.6). ∎

Lemma 3.1 shows that every member of $\mathcal{C}_\theta$ has the effective penalty $\rho + s$, whatever the center stepsizes are. The center stepsize β only places the dual center between two classical choices, and α decides whether the primal variable is smoothed.

Zero center ($\beta = 0$). Then $\boldsymbol{\mu}^k = \mathbf{0}$, and $\mathcal{D}_{\rho+s}(\boldsymbol{x}, \mathbf{0}) = f(\boldsymbol{x}) + \min_{\boldsymbol{y}}\{h(\boldsymbol{y}) + \frac{\rho+s}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2\}$ is the quadratic penalty function with parameter $\rho + s$. The dual step carries the perturbation $-\boldsymbol{\lambda}^k/s$, as in perturbed GDA [21, 35] and perturbed smoothed GDA [14].

Current multiplier as center ($\beta = 1$). Then $\boldsymbol{\mu}^k = \boldsymbol{\lambda}^k$, the gradient of the dual regularizer vanishes at every dual step, and

$$\boldsymbol{\lambda}^{k+1} = \Big(1 - \frac{\tau}{\rho}\Big)\boldsymbol{\lambda}^k + \frac{\tau}{\rho}\,\boldsymbol{q}^k, \qquad \boldsymbol{q}^k = \mathrm{prox}_{\rho h^*}\big(\boldsymbol{\lambda}^k + \rho\mathcal{A}(\boldsymbol{x}^{k+1})\big).$$

For $\tau = \rho$, this is the multiplier update of the augmented Lagrangian method [11, 27, 29], that is, a proximal ascent step on $\mathcal{L}(\boldsymbol{x}^{k+1}, \cdot)$ by (3.5). The iterates do not depend on s, so this choice also reproduces the case without dual regularizer, $s = \infty$.

Moving center ($0 < \beta < 1$). The dual center is an exponentially weighted average of past multipliers, and the method has the structure of doubly smoothed GDA [39].

Table 2 collects these cases. The identities $\boldsymbol{\mu}^k = \mathbf{0}$, $\boldsymbol{\mu}^k = \boldsymbol{\lambda}^k$, and $\boldsymbol{z}^k = \boldsymbol{x}^k$ for $\beta = 0$, $\beta = 1$, and $\alpha = 1$, respectively, follow by induction from the zero initialization. Every entry refers to the cited update structure applied to $\mathcal{D}_\rho$ with $\rho > 0$; methods acting on $\mathcal{L}$ itself are compared in Section 5.1.

4. Main Results.

4.1. Main result. We use the problem class of Definition 2.1, the family $\mathcal{C}_\theta$ of Definition 3.2, and the complexity of Definition 3.3, with the normalized constants $\mathbf{C}_0 := (1, 1, 0, 1, 1)$. Since $L_{\mathcal{A}} = 0$, every instance in $\mathcal{I}(\mathbf{C}_0, \Delta)$ has an affine inner mapping.

We now present the iteration complexity of $\mathcal{C}_\theta$ under the choice $\beta = 1$. We choose this setting for two reasons. First, when $\beta = 1$, SPLALM reduces to Smoothed GDA applied to $\mathcal{D}_\rho$ (Table 2), whose iteration complexity is known. Second, Theorem 4.2 below holds for any arbitrary choice of $(\alpha, \beta) \in [0,1]^2$, including $\beta = 1$. Stating the upper bound under the same setting yields a direct tightness guarantee.

THEOREM 4.1 (Iteration complexity of SPLALM with $\beta = 1$). *There exist $\bar{\gamma} > 0$, depending only on $\mathbf{C}$ and ρ, and $\bar{\tau} > 0$, depending only on $\mathbf{C}$, ρ, and η/γ, such that the following holds. Let $\gamma \le \bar{\gamma}$, $\eta \le \gamma/2$, $\tau \le \bar{\tau}$, and $s \ge \tau$. Then there exist $c > 0$, depending only on θ and $\mathbf{C}$, and $C > 0$, depending only on θ, $\mathbf{C}$, and Δ, such that, given any $\epsilon \in (0,1]$ and $\alpha := c\,\epsilon^2$,*

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, 1; I) \le C\,\epsilon^{-4} \qquad \text{for all } I \in \mathcal{I}(\mathbf{C}, \Delta).$$

The proof is given in Appendix B.

Remark 4.1. (i) When $\beta = 1$, SPLALM reduces to Smoothed GDA applied to $\mathcal{D}_\rho$. A result analogous to Theorem 4.1 has been obtained in [37, Theorem 3.4] and [14, Theorem B.1]. However, the initial gap there depends on the initialization, the dual domain is a given compact set, and the stationarity measure is evaluated at $(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1})$. Theorem 4.1 bounds the initial gap by the constants of $\mathcal{I}(\mathbf{C}, \Delta)$, keeps the multipliers in the interior of a ball without any projection, and evaluates the KKT residual at $(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k})$, the point that SPLALM outputs. (ii) The iterates do not depend on s when $\beta = 1$. The upper bound requires only one member of $\mathcal{C}_\theta$; for $\beta = 0$, the instance in Section 5.4 shows that no upper bound holds when $\epsilon(\rho + s) \le c_1$.

THEOREM 4.2 (Lower bound uniform in ρ and s). *Fix $0 < \eta \le \gamma$, $\tau > 0$, and $\Delta > 0$. There exist constants $c_1 = c_1(\gamma, \eta, \tau) > 0$ and $c_2 = c_2(\gamma, \tau, \Delta) > 0$, independent of ρ, s, α, and β, such that the following holds. For every $\epsilon \in (0, c_2)$, every $\rho > 0$ and $s \ge \tau$ with $\epsilon(\rho + s) \le c_1$, and every $(\alpha, \beta) \in [0,1]^2$, there exists an instance $I \in \mathcal{I}(\mathbf{C}_0, \Delta)$ on which $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ requires at least*

$$\frac{\Delta\,\epsilon^{-4}}{16384\,\gamma \max\{1, 4\tau, 4\gamma\tau\}}$$

iterations before one of its outputs is an ϵ-KKT point. Equivalently,

$$T_\epsilon^{\mathrm{KKT}}(\mathcal{C}_\theta, \mathcal{I}(\mathbf{C}_0, \Delta)) \ge \Delta\epsilon^{-4}/(16384\,\gamma \max\{1, 4\tau, 4\gamma\tau\})$$

.

Explicit choices of c_1 and c_2 are given in (A.4). **TODO:** State the GS version T_ϵ^{GS} after the strict-convex perturbation lemma is proved.

COROLLARY 4.1 (Necessary scale of the effective penalty). *Let $\rho = \rho(\epsilon) > 0$ and $s = s(\epsilon) \ge \tau$. If $\rho + s = o(\epsilon^{-1})$, then Theorem 4.2 applies for all sufficiently small ϵ, and the complexity of $\mathcal{C}_\theta$ is $\Omega(\epsilon^{-4})$. In particular, an $O(\epsilon^{-3})$ guarantee for $\mathcal{C}_\theta$ requires $\rho(\epsilon) + s(\epsilon) > c_1/\epsilon$ for all sufficiently small ϵ.*

5. Hard Instances and Proof of Theorem 4.2. In Section 4, we established a lower bound on the iteration complexity of the SPLALM family that holds for every choice of constant center stepsizes. In this section, we prove this result by constructing hard instances.

Fix ϵ and θ satisfying the conditions of Theorem 4.2. The lower bound has the quantified form

$$\forall\,(\alpha, \beta) \in [0,1]^2, \quad \exists\, I \in \mathcal{I}(\mathbf{C}_0, \Delta): \quad T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) \ge \Omega(\Delta\epsilon^{-4}).$$

Thus, different choices of (α,β) may require different hard instances. For each pair, we construct an instance in $\mathcal{I}(\mathbf{C}_0,\Delta)$ on which $\mathrm{SPLALM}_{\theta,\alpha,\beta}$, started from the zero initialization prescribed in Definition 3.1, requires at least the stated number of iterations to produce an ϵ-KKT output.

We split $[0,1]^2$ into three parts by the threshold

$$\alpha_\epsilon := 1024\max\{1,4\tau,4\gamma\tau\}\,\epsilon^2,$$

namely

$$\begin{aligned}[0,1]^2 = &\big\{(\alpha,\beta)\in[0,1]^2 : \alpha\le\alpha_\epsilon\big\}\\ &\cup\big\{(\alpha,\beta)\in[0,1]^2 : \alpha>\alpha_\epsilon,\ \beta>0\big\}\\ &\cup\big\{(\alpha,\beta)\in[0,1]^2 : \alpha>\alpha_\epsilon,\ \beta=0\big\},\end{aligned}$$

and use a separate hard instance for each part. For $\alpha\le\alpha_\epsilon$, the bound obtained in Section 5.2 is of order $\Delta\epsilon^{-4}$. This threshold ensures the parameter estimates required by the rotating instance in Section 5.3.

5.1. Relation to Smoothed GDA and DS-GDA. For the outputs of $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ in Definition 3.1, we write $R_k := \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{k+1},\boldsymbol{y}^{k+1},\boldsymbol{q}^k)$ for $k\ge 0$.

$\mathrm{SPLALM}_{\theta,\alpha,\beta}$ in Definition 3.1 uses $\mathrm{prox}_{h/\rho}$, so it requires $\rho>0$. Smoothed GDA and DS-GDA contain no such term. They nevertheless generate the same iterates as $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ as long as $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})=\mathbf{0}$. To state this, define

$$\mathcal{R}_\rho := \big\{(\boldsymbol{x},\boldsymbol{\lambda}):\ \boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})\in\partial h(\mathbf{0})\big\},\qquad \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}) := f(\boldsymbol{x})+\frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2+\langle\boldsymbol{\lambda},\mathcal{A}(\boldsymbol{x})\rangle.$$

Proposition 5.1 (Reduction on $\mathcal{R}_\rho$). *Let $\rho>0$ and $(\boldsymbol{x},\boldsymbol{\lambda})\in\mathcal{R}_\rho$. Then $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})=\mathbf{0}$, $\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})=\boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})$, and*

$$\nabla f(\boldsymbol{x})+J_{\mathcal{A}}(\boldsymbol{x})^\top\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})=\nabla_{\boldsymbol{x}}\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}),\qquad \mathcal{A}(\boldsymbol{x})-\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})=\nabla_{\boldsymbol{\lambda}}\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}).$$

Moreover, $\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})=\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda})+h(\mathbf{0})$.

Proof. Set $\boldsymbol{v}:=\mathcal{A}(\boldsymbol{x})+\boldsymbol{\lambda}/\rho$. By (3.2), $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})$ is the unique minimizer of $h(\boldsymbol{y})+\frac{\rho}{2}\|\boldsymbol{y}-\boldsymbol{v}\|^2$, and $\boldsymbol{y}$ is this minimizer if and only if $\rho(\boldsymbol{v}-\boldsymbol{y})\in\partial h(\boldsymbol{y})$. For $\boldsymbol{y}=\mathbf{0}$ the inclusion reads $\boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})\in\partial h(\mathbf{0})$, which holds because $(\boldsymbol{x},\boldsymbol{\lambda})\in\mathcal{R}_\rho$. Hence $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})=\mathbf{0}$ and $\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})=\boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})$. Since $\nabla_{\boldsymbol{x}}\big(\frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2\big)=\rho J_{\mathcal{A}}(\boldsymbol{x})^\top\mathcal{A}(\boldsymbol{x})$,

$$\nabla f(\boldsymbol{x})+J_{\mathcal{A}}(\boldsymbol{x})^\top\big(\boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})\big)=\nabla_{\boldsymbol{x}}\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}),$$

and $\mathcal{A}(\boldsymbol{x})-\mathbf{0}=\nabla_{\boldsymbol{\lambda}}\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda})$. Finally, completing the square in (3.1) gives $\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})=f(\boldsymbol{x})+\min_{\boldsymbol{y}}\{h(\boldsymbol{y})+\frac{\rho}{2}\|\boldsymbol{y}-\boldsymbol{v}\|^2\}-\frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2$, and the minimum equals $h(\mathbf{0})+\frac{\rho}{2}\|\boldsymbol{v}\|^2$ because its minimizer is $\mathbf{0}$. Hence

$$\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})=f(\boldsymbol{x})+h(\mathbf{0})+\frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2+\langle\boldsymbol{\lambda},\mathcal{A}(\boldsymbol{x})\rangle+\frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2-\frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2=\Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda})+h(\mathbf{0}).$$

∎

For $\theta=(\rho,s,\gamma,\eta,\tau)$ and $(\alpha,\beta)\in[0,1]^2$, consider the recursion

$$\begin{aligned}\boldsymbol{x}^{k+1}&=\boldsymbol{x}^k-\eta\Big[\nabla_{\boldsymbol{x}}\Psi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)+\frac{1}{\gamma}(\boldsymbol{x}^k-\boldsymbol{z}^k)\Big],\\ \boldsymbol{\lambda}^{k+1}&=\boldsymbol{\lambda}^k+\tau\Big[\nabla_{\boldsymbol{\lambda}}\Psi_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k)-\frac{1}{s}(\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k)\Big],\\ \boldsymbol{z}^{k+1}&=\boldsymbol{z}^k+\alpha(\boldsymbol{x}^{k+1}-\boldsymbol{z}^k),\qquad \boldsymbol{\mu}^{k+1}=\boldsymbol{\mu}^k+\beta(\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^k).\end{aligned}\tag{5.1}$$

PROPOSITION 5.2 (Embedding). *Let $\rho > 0$, and let the sequence $\boldsymbol{w}^k := (\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k, \boldsymbol{\mu}^k)$ be generated by* SPLALM$_{\theta,\alpha,\beta}$. *Suppose that for some $K \geq 1$, $(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) \in \mathcal{R}_\rho$ and $(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k) \in \mathcal{R}_\rho$ for all $0 \leq k < K$. Then (5.1) holds for $0 \leq k < K$, and*

$$R_k = \max\big\{\|\nabla_{\boldsymbol{x}} \Psi_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k)\|,\ \|\nabla_{\boldsymbol{\lambda}} \Psi_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k)\|\big\}, \qquad 0 \leq k < K.$$

Proof. Apply Proposition 5.1 at $(\boldsymbol{x}^k, \boldsymbol{\lambda}^k)$ in the $\boldsymbol{x}$-update of (3.4), and at $(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k)$ in the $\boldsymbol{\lambda}$-update. The same point gives $\boldsymbol{y}^{k+1} = \boldsymbol{0}$ and $\boldsymbol{q}^k = \boldsymbol{\lambda}^k + \rho \mathcal{A}(\boldsymbol{x}^{k+1})$, so the formula for R_k follows from (2.2). ∎

Recursion (5.1) is the update of DS-GDA [39] for the saddle function Ψ_ρ, with the same order of updates; the parameter correspondence for DS-GDA and its special cases is listed in Table 2.

Remark 5.1. The following points limit the scope of Propositions 5.1 and 5.2.

(i) The projections onto X and Y in the cited methods are inactive when the trajectory stays in the interior of $X \times Y$. For $h = \|\cdot\|_2$, we have $\partial h(\boldsymbol{0}) = \mathbb{B}(\boldsymbol{0}, 1)$ and $h^* = \iota_{\mathbb{B}(\boldsymbol{0},1)}$, so $Y = \mathbb{B}(\boldsymbol{0}, 1)$ is the natural dual set.

(ii) For a smooth $\tilde{f}$, SPLALM$_{\theta,\alpha,\beta}$ on the instance $\big(\tilde{f} - \frac{\rho}{2}\|\mathcal{A}\|^2, \mathcal{A}, h\big)$ generates (5.1) with $\Psi = \tilde{f} + \langle \cdot, \mathcal{A} \rangle$ whenever its trajectory stays in $\mathcal{R}_\rho$. A hard instance of this form for Smoothed GDA or DS-GDA is therefore a hard instance for SPLALM$_{\theta,\alpha,\beta}$ with any $\rho > 0$, provided that $\tilde{f} - \frac{\rho}{2}\|\mathcal{A}\|^2$ satisfies the bounds in Definition 2.1.

(iii) Outside $\mathcal{R}_\rho$, the two recursions differ. The analysis of SPLALM uses the smoothness of $\mathcal{D}_\rho(\cdot, \boldsymbol{\lambda})$ in (3.1), which requires $\rho > 0$; it does not apply to GDA-type methods.

(iv) The case $s = \infty$ is reproduced by $\beta = 1$ for every finite s; see Section 3.2. PSGDA with $r_y = \Theta(\epsilon)$ has $s = \Theta(\epsilon^{-1})$; whether it satisfies $\epsilon(\rho + s) \leq c_1$ depends on the constants in r_y and c_1.

TODO: State the transfer of Theorem 4.2 to Smoothed GDA and DS-GDA as a corollary.

5.2. Small primal center stepsize α. In this subsection, we treat the pairs (α, β) with $\alpha \leq \alpha_\epsilon$. We construct a hard instance on which, for every $\beta \in [0, 1]$, SPLALM$_{\theta,\alpha,\beta}$ requires more than $\Delta/(16\gamma\alpha\epsilon^2)$ iterations to produce an ϵ-KKT output. The hard instance is given below.

Example 5.1. Let $0 < \epsilon \leq \min\{1/2, \sqrt{\Delta}/2\}$ and $x_\star := \Delta/(2\epsilon) + \epsilon$. Define $\mathcal{A} \equiv 0$, $h(y) = |y|$, and $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 2\epsilon(x_\star - x) - 2\epsilon^2, & x < x_\star - 2\epsilon, \\ \dfrac{1}{2}(x - x_\star)^2, & |x - x_\star| \leq 2\epsilon, \\ 2\epsilon(x - x_\star) - 2\epsilon^2, & x > x_\star + 2\epsilon. \end{cases}$$

Then f is 1-smooth with $\|\nabla f\| \leq 2\epsilon \leq 1$, $\mathcal{A}$ satisfies $L_{\mathcal{A}} = G_{\mathcal{A}} = 0$, and h is convex and 1-Lipschitz. Since $\epsilon \leq \sqrt{\Delta}/2$, we have $x_\star \geq 2\epsilon$, thus $f(0) = 2\epsilon x_\star - 2\epsilon^2 = \Delta$. Moreover, $\inf f = f(x_\star) = 0$. Hence $(f, \mathcal{A}, h) \in \mathcal{I}(\mathbf{C}_0, \Delta)$.

Remark 5.2. The construction ensures that $\inf f = 0$ and $f(0) = \Delta$, so the initial objective gap is exactly Δ. This allows us to make the dependence of the iteration lower bound on Δ explicit.

This instance shows that a small primal center stepsize can force $\Omega\big(\Delta/(\gamma\alpha\epsilon^2)\big)$ iterations even on a smooth one-dimensional problem, regardless of the dual center stepsize. The iterates must traverse a long constant-gradient interval before an output can satisfy the KKT tolerance. A small α limits how quickly the center moves, and the primal variable stays within a bounded distance of that center. Since

the multipliers remain zero, every output on this interval has KKT residual $2\epsilon > \epsilon$. The next lemma bounds the distance traveled and converts it into a lower bound on the number of iterations.

LEMMA 5.1. *Let $(\alpha, \beta) \in [0,1]^2$, and suppose that ϵ satisfies the condition in Example 5.1 and $\epsilon^2 \le \Delta/(16\gamma)$. Run* SPLALM$_{\theta,\alpha,\beta}$ *on Example 5.1. Then the following hold.*

(i) For all $k \ge 0$, $\lambda^k = \mu^k = 0$, $y^{k+1} = 0$, and $R_k = \|\nabla f(x^{k+1})\|$.

(ii) If $\alpha > 0$, then for every integer k with $0 \le k \le \Delta/(16\gamma\alpha\epsilon^2)$,

$$\|z^k - z^0\| \le 2\alpha\gamma\epsilon\, k, \qquad \|x^k - z^k\| \le 2\gamma\epsilon,$$

and

$$\|x^k - x^0\| \le 2\gamma\epsilon(1 + \alpha k) \le x_\star - 2\epsilon.$$

When $\alpha = 0$, these bounds hold for all $k \ge 0$.

(iii) Let I denote the instance of Example 5.1. Then $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) > \Delta/(16\gamma\alpha\epsilon^2)$ if $\alpha > 0$, and $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) = +\infty$ if $\alpha = 0$.

Proof. (i) Since $\mathcal{A} \equiv 0$ and $\partial h(0) = [-1, 1]$, every pair $(x, 0)$ lies in $\mathcal{R}_\rho$, and $\Psi_\rho(x, \lambda) = f(x)$. The λ-update in (5.1) reads $\lambda^{k+1} = \lambda^k - \frac{\tau}{s}(\lambda^k - \mu^k)$. Starting from $\lambda^0 = \mu^0 = 0$, induction gives $\lambda^k = \mu^k = 0$ for all k. Proposition 5.2 then gives $y^{k+1} = 0$ and $R_k = \max\{\|\nabla f(x^{k+1})\|, 0\}$.

(ii) We use the Euclidean norm, which coincides with the absolute value in one dimension. By (i), the primal update gives

$$x^{k+1} - z^k = \left(1 - \frac{\eta}{\gamma}\right)(x^k - z^k) - \eta\nabla f(x^k).$$

Since $\|\nabla f(x)\| \le 2\epsilon$ for every x and $0 < \eta \le \gamma$,

$$\|x^{k+1} - z^k\| \le \left(1 - \frac{\eta}{\gamma}\right)\|x^k - z^k\| + 2\eta\epsilon.$$

The center update implies

$$\|x^{k+1} - z^{k+1}\| = (1 - \alpha)\|x^{k+1} - z^k\|, \qquad \|z^{k+1} - z^k\| = \alpha\|x^{k+1} - z^k\|.$$

Starting from $x^0 = z^0$, induction gives $\|x^k - z^k\| \le 2\gamma\epsilon$ and $\|x^{k+1} - z^k\| \le 2\gamma\epsilon$ for all $k \ge 0$. Consequently,

$$\|z^k - z^0\| \le \sum_{j=0}^{k-1} \|z^{j+1} - z^j\| \le 2\alpha\gamma\epsilon k.$$

Using $x^0 = z^0 = 0$, we obtain

$$\|x^k - x^0\| \le \|x^k - z^k\| + \|z^k - z^0\| \le 2\gamma\epsilon(1 + \alpha k).$$

If $\alpha > 0$, then for every integer $k \le \Delta/(16\gamma\alpha\epsilon^2)$, the assumptions $\epsilon^2 \le \Delta/(16\gamma)$ and $\epsilon^2 \le \Delta/4$ imply

$$\|x^k - x^0\| \le 2\gamma\epsilon + \frac{\Delta}{8\epsilon} \le \frac{\Delta}{4\epsilon} \le \frac{\Delta}{2\epsilon} - \epsilon = x_\star - 2\epsilon.$$

In particular, $x^k \le \|x^k - x^0\| \le x_\star - 2\epsilon$, so the iterate lies in the constant-gradient region. When $\alpha = 0$, the center remains at $z^k = z^0 = 0$, and the same conclusion holds for every $k \ge 0$.

(iii) Suppose $\alpha > 0$. By (ii), every output with $k+1 \le \Delta/(16\gamma\alpha\epsilon^2)$ satisfies $x^{k+1} \le x_\star - 2\epsilon$. By (i), its residual is $R_k = \|\nabla f(x^{k+1})\| = 2\epsilon > \epsilon$. Thus, none of the first $\lfloor \Delta/(16\gamma\alpha\epsilon^2) \rfloor$ outputs is an ϵ-KKT point. Definition 3.3 yields

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) \ge \left\lfloor \frac{\Delta}{16\gamma\alpha\epsilon^2} \right\rfloor + 1 > \frac{\Delta}{16\gamma\alpha\epsilon^2}.$$

When $\alpha = 0$, (ii) keeps every iterate in the constant-gradient region. Hence $R_k = 2\epsilon > \epsilon$ for all $k \ge 0$, and $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) = +\infty$. ∎

For $0 < \alpha \le \alpha_\epsilon$, Lemma 5.1 (iii) gives

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) > \frac{\Delta}{16\gamma\alpha\epsilon^2} \ge \frac{\Delta\, \epsilon^{-4}}{16384\, \gamma \max\{1, 4\tau, 4\gamma\tau\}}.$$

5.3. Large primal center stepsize α, and $\beta > 0$. In this subsection, we prove the lower bound for $\alpha > \alpha_\epsilon$ and $\beta > 0$. For every such pair, we construct a hard instance on which SPLALM, initialized at zero, never produces an ϵ-KKT point.

Motivated by the quadratic hard instances in [14, Example 5.1] and [22, Appendix C.1], we construct the following instance.

Example 5.2. Let $b, \kappa, X > 0$ and $x_c, \lambda_c \in \mathbb{C}$ be given in Appendix A.1, and set $R_* := \frac{5}{4}X$. Define $\mathcal{A}(x) = b(x - x_c)$, $h(y) = \|y\|_2$, and

$$f(x) = \psi\big(\|x - x_c\|\big) - b\langle \lambda_c, x - x_c\rangle, \qquad \psi(r) = \begin{cases} -\dfrac{\kappa}{2} r^2, & 0 \le r \le R_*, \\ \dfrac{\kappa}{2}(r - 2R_*)^2 - \kappa R_*^2, & R_* \le r \le 3R_*, \\ \kappa R_*(r - 3R_*) - \dfrac{\kappa}{2} R_*^2, & r \ge 3R_*. \end{cases}$$

To establish this lower bound, we analyze the SPLALM updates on Example 5.2. Specifically, since $h = \|\cdot\|_2$, Proposition 5.1 ensures that both proximal outputs vanish whenever

$$\max_{j \in \{k, k+1\}} \|\lambda^k + \rho \mathcal{A}(x^j)\| \le 1.$$

Under this condition, SPLALM follows (5.1). In the regime where $\|x - x_c\| \le R_*$ and $\|\lambda + \rho\mathcal{A}(x)\| \le 1$, the reduced function takes the quadratic form

$$\Psi_\rho(x, \lambda) = -\frac{a}{2}\|x - x_c\|^2 + b\langle \lambda - \lambda_c, x - x_c\rangle.$$

Here $a := \kappa - \rho b^2$. The gradients of this quadratic function are linear in $x - x_c$ and $\lambda - \lambda_c$. Therefore, let define the iteration sequence $\boldsymbol{w}^k := (x^k, \lambda^k, z^k, \mu^k)$ and $\boldsymbol{w}_c := (x_c, \lambda_c, x_c, \lambda_c)$. Whenever the proximal condition above and $\|x^k - x_c\| \le R_*$ hold, the four updates take the linear form

$$\boldsymbol{w}^{k+1} - \boldsymbol{w}_c = \mathbb{T}(\boldsymbol{w}^k - \boldsymbol{w}_c),$$

with $\mathbb{T}$ defined as

$$\mathbb{T} = \begin{pmatrix} 1 + \eta a - \frac{\eta}{\gamma} & -\eta b & \frac{\eta}{\gamma} & 0 \\ \tau b\big(1 + \eta a - \frac{\eta}{\gamma}\big) & 1 - \frac{\tau}{s} - \eta\tau b^2 & \frac{\eta\tau b}{\gamma} & \frac{\tau}{s} \\ \alpha\big(1 + \eta a - \frac{\eta}{\gamma}\big) & -\alpha\eta b & 1 - \alpha + \frac{\alpha\eta}{\gamma} & 0 \\ \beta\tau b\big(1 + \eta a - \frac{\eta}{\gamma}\big) & \beta\big(1 - \frac{\tau}{s} - \eta\tau b^2\big) & \frac{\beta\eta\tau b}{\gamma} & 1 - \beta + \frac{\beta\tau}{s} \end{pmatrix}. \tag{5.2}$$

Under the conditions above, analyzing the SPLALM iterates reduces to studying the spectral properties of $\mathbb{T}$. The core idea is to construct a bounded trajectory whose KKT residual remains above ϵ. We choose a and b so that $\mathbb{T}$ has a non-real eigenvalue of modulus one, which gives a component that does not decay.

The construction in [14, Section 5.1.1] chooses both the hard instance and the initial point, with the latter lying along an eigenvector of the iteration matrix. In our setting, the initial point is fixed at zero, and we choose the instance alone. We therefore need to obtain the required trajectory from this prescribed initial point. To this end, we choose the centers x_c and λ_c in the instance so that the shifted initial state is the sum of an eigenvector of $\mathbb{T}$ and a decaying component. The following lemma gives the spectral properties and this decomposition for the parameters defined in Appendix A.1.

LEMMA 5.2. *Under the conditions of Lemma 5.3, let $\mathbb{T}$ be the matrix in (5.2). Then:*
(i) $\mathbb{T}\boldsymbol{v} = \zeta\boldsymbol{v}$, where $|\zeta| = 1$ and $\zeta \notin \mathbb{R}$;
(ii) $\boldsymbol{r}$ lies in a $\mathbb{T}$-invariant subspace on which the spectral radius of $\mathbb{T}$ is less than one;
(iii) $\boldsymbol{v} + \boldsymbol{r} = -\boldsymbol{w}_c$.

By Lemma 5.2, the linear recursion initialized at zero has the explicit solution

$$\boldsymbol{w}^k - \boldsymbol{w}_c = \zeta^k \boldsymbol{v} + \mathbb{T}^k \boldsymbol{r}.$$

We next bound $\mathbb{T}^k\boldsymbol{r}$ uniformly in k to verify that this sequence stays in the quadratic region, both proximal outputs vanish, and the KKT residual remains above ϵ.

LEMMA 5.3. *Let $\alpha > \alpha_\epsilon$, $0 < \beta \le 1$, $0 < \epsilon < c_2$, and $\epsilon(\rho + s) \le c_1$, with c_1, c_2 as specified in Theorem 4.2. Then:*
(i) the instance of Example 5.2 belongs to $\mathcal{I}(\mathbf{C}_0, \Delta)$;
(ii) for all $k \ge 0$,

$$\|(\mathbb{T}^k\boldsymbol{r})_x\| \le \frac{X}{4}, \qquad \|(\mathbb{T}^k\boldsymbol{r})_\lambda\| \le s\Big(\frac{\gamma}{\eta} + \frac{5}{4}\Big)bX.$$

The proofs of Lemmas 5.2 and 5.3 are given in Appendix A.

Using these estimates, we now show that the sequence generated by the linear recursion coincides with the SPLALM iterates and that its KKT residual remains above ϵ at every iteration.

LEMMA 5.4. *Under the conditions of Lemma 5.3, the iterates of* SPLALM$_{\theta,\alpha,\beta}$ *on Example 5.2 satisfy, for all $k \ge 0$,*

$$x^k - x_c = \zeta^k X + (\mathbb{T}^k\boldsymbol{r})_x, \qquad \lambda^k - \lambda_c = \zeta^k \Lambda + (\mathbb{T}^k\boldsymbol{r})_\lambda,$$

and $R_k \ge \frac{7}{6}\epsilon$. Hence $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) = +\infty$.

Proof. We prove by induction that, for all $k \ge 0$,

$$\boldsymbol{w}^k - \boldsymbol{w}_c = \zeta^k \boldsymbol{v} + \mathbb{T}^k \boldsymbol{r}.$$

Base case $k = 0$. By Lemma 5.2 (iii), $\boldsymbol{v} + \boldsymbol{r} = -\boldsymbol{w}_c$. Since $\boldsymbol{w}^0 = \mathbf{0}$, the identity holds for $k = 0$.

Inductive step. Assume that the identity holds for some $k \ge 0$. Since $|\zeta| = 1$, Lemma 5.3 (ii) gives

$$\frac{3}{4}X \le \|x^k - x_c\| \le \frac{5}{4}X = R_*.$$

Thus, x^k lies in the quadratic region. Moreover, the induction hypothesis and Lemma 5.2 (iii) imply $\|\lambda^k - \lambda_c\| \le \|\Lambda\| + \|(\mathbb{T}^k\boldsymbol{r})_\lambda\|$ and $\|\lambda_c\| \le \|\Lambda\| + \|\boldsymbol{r}_\lambda\|$. Lemma 5.3 (ii) and the parameter bounds in

Lemma A.1 therefore yield

$$\|\lambda^k + \rho\mathcal{A}(x^k)\| \le 2\|\Lambda\| + 2s\Big(\frac{\gamma}{\eta} + \frac{5}{4}\Big)bX + \frac{5}{4}\rho bX \le \frac{1}{3}\Big(2 + \frac{1}{4}\Big) = \frac{3}{4}.$$

Hence $(x^k, \lambda^k) \in \mathcal{R}_\rho$. By Proposition 5.1, the primal update is given by the first row of (5.2). Using the induction hypothesis and $\mathbb{T}\boldsymbol{v} = \zeta\boldsymbol{v}$, we obtain

$$x^{k+1} - x_c = \zeta^{k+1}X + (\mathbb{T}^{k+1}\boldsymbol{r})_x.$$

Lemma 5.3 (ii) then gives $\frac{3}{4}X \le \|x^{k+1} - x_c\| \le \frac{5}{4}X$. The same estimate as above, with x^{k+1} in place of x^k, yields

$$\|\lambda^k + \rho\mathcal{A}(x^{k+1})\| \le \frac{3}{4}.$$

Thus, $(x^{k+1}, \lambda^k) \in \mathcal{R}_\rho$ as well, so Proposition 5.1 also applies to the dual update. Together with the center updates, this gives

$$\boldsymbol{w}^{k+1} - \boldsymbol{w}_c = \mathbb{T}(\boldsymbol{w}^k - \boldsymbol{w}_c) = \zeta^{k+1}\boldsymbol{v} + \mathbb{T}^{k+1}\boldsymbol{r}.$$

This completes the induction. In particular, both proximal outputs vanish at every iteration, and all primal iterates lie in the prescribed annulus.

Finally, we bound the residual. By Proposition 5.2,

$$R_k = \max\big\{\|-a(x^{k+1} - x_c) + b(\lambda^k - \lambda_c)\|,\ b\|x^{k+1} - x_c\|\big\}.$$

The second term is at least $\frac{3}{4}bX$. For the first term, Lemma 5.3 (ii) and Lemma A.1 give

$$\|-a(x^{k+1} - x_c) + b(\lambda^k - \lambda_c)\| \ge b\|\Lambda\| - s\Big(\frac{\gamma}{\eta} + \frac{5}{4}\Big)b^2X - \frac{5}{4}aX \ge \frac{27}{32}b\|\Lambda\| - s\Big(\frac{\gamma}{\eta} + \frac{5}{2}\Big)b^2X.$$

If $\|\Lambda\| \ge X$, Lemma A.1 bounds the right-hand side below by $\frac{23}{32}b\|\Lambda\|$. If $X \ge \|\Lambda\|$, we use the second term instead. In both cases, $R_k \ge \frac{b}{2}\max\{X, \|\Lambda\|\} = \frac{b}{6} \ge \frac{7}{6}\epsilon$. ∎

5.4. Large primal center stepsize α, and $\beta = 0$. In this subsection, we prove the lower bound for $\alpha > \alpha_\epsilon$ and $\beta = 0$. The instance below keeps the objective of Example 5.1 and changes the inner mapping and the outer function. On this instance, SPLALM$_{\theta,\alpha,0}$, initialized at zero, never produces an ϵ-KKT point, for every $\alpha \in [0, 1]$.

Example 5.3. Let ϵ satisfy the condition in Example 5.1, and let f be the function defined there. Set

$$q_h := 2s + 1, \qquad a_0 := \frac{4\epsilon(\rho + 3s + 1)}{q_h}.$$

Define $\mathcal{A} \equiv a_0$ and $h : \mathbb{R} \to \mathbb{R}$ by

$$h(y) = \begin{cases} \dfrac{q_h}{2}y^2, & |y| \le q_h^{-1}, \\ |y| - \dfrac{1}{2q_h}, & |y| > q_h^{-1}. \end{cases}$$

Then h is convex and 1-Lipschitz, and $\mathcal{A}$ satisfies $L_\mathcal{A} = G_\mathcal{A} = 0$. Since $F = f + h(a_0)$ differs from f by a constant, Remark 5.2 gives $F(0) - \inf F = f(0) - \inf f = \Delta$. Together with the properties of f established in Example 5.1, this gives $(f, \mathcal{A}, h) \in \mathcal{I}(\mathbf{C}_0, \Delta)$.

When $\beta = 0$, the dual center stays at $\mu^k = \mu^0 = 0$. The dual perturbation term $-\frac{1}{2s}\|\lambda\|^2$ in (3.3) therefore pulls λ^k toward zero at every iteration. On this instance, the output y^{k+1} equals a_0 only when $\lambda^k = \nabla h(a_0) = q_h a_0$, whereas the dual iterates converge to a smaller value. The difference keeps the feasibility residual $\|\mathcal{A}(x^{k+1}) - y^{k+1}\|$ above ϵ at every iteration. Since $J_{\mathcal{A}} = 0$, the multiplier does not enter the primal update, and the dual iterates have the closed form given in the next lemma.

LEMMA 5.5. *Let $\alpha \in [0,1]$ and $\beta = 0$. Suppose that ϵ satisfies the condition in Example 5.1 and $\epsilon(\rho + s) \le c_1$, with c_1 as specified in Theorem 4.2. Run* SPLALM$_{\theta,\alpha,0}$ *on Example 5.3, and set*

$$\varphi := 1 - \frac{\tau}{s} - \frac{\tau}{\rho + q_h}, \qquad \bar{\lambda} := \frac{s\, q_h a_0}{\rho + q_h + s}.$$

Then the following hold.

(i) For all $k \ge 0$,

$$\mu^k = 0, \qquad \lambda^k = \bar{\lambda}\big(1 - \varphi^k\big), \qquad y^{k+1} = \frac{\rho a_0 + \lambda^k}{\rho + q_h}.$$

(ii) For all $k \ge 0$, $R_k \ge \|\mathcal{A}(x^{k+1}) - y^{k+1}\| \ge 3\epsilon$. Hence, with I the instance of Example 5.3, $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, 0; I) = +\infty$.

Proof. We first collect three bounds on the parameters. Since $0 < \tau \le s$ and $\rho + q_h > 2s$,

$$-\frac{1}{2} < -\frac{\tau}{\rho + q_h} \le \varphi < 1, \qquad \frac{s}{\rho + q_h} < \frac{1}{2}. \tag{5.3}$$

Hence $\varphi^k \ge -\frac{1}{2}$ and $0 \le 1 - \varphi^k \le 2$ for all $k \ge 0$. Since $2s \le \rho + q_h + s$, it follows that $0 \le \bar{\lambda}(1 - \varphi^k) \le 2\bar{\lambda} \le q_h a_0$. Finally, $s \ge \tau$ gives $1 \le (\rho + s)/\tau$, so the third term in the definition of c_1 yields

$$q_h a_0 = 4\epsilon(\rho + 3s + 1) \le 4\epsilon\big(3 + \tau^{-1}\big)(\rho + s) \le 4c_1\big(3 + \tau^{-1}\big) \le \frac{1}{4}.$$

(i) Since $\beta = 0$ and $\mu^0 = 0$, the dual center update gives $\mu^k = 0$ for all k. Next, fix $x \in \mathbb{R}$ and $\lambda \in [0, q_h a_0]$, and set $y_\lambda := (\rho a_0 + \lambda)/(\rho + q_h)$. Then $0 \le y_\lambda \le a_0 \le q_h^{-1}$, where the last inequality uses $q_h a_0 \le \frac{1}{4}$. On $[-q_h^{-1}, q_h^{-1}]$, the function h is differentiable with $\nabla h(y) = q_h y$, and y_λ satisfies

$$q_h y_\lambda - \lambda - \rho\big(a_0 - y_\lambda\big) = 0,$$

which is the optimality condition of the strongly convex problem in (3.1). Hence $y(x, \lambda) = y_\lambda$, and

$$\mathcal{A}(x) - y(x, \lambda) = \frac{q_h a_0 - \lambda}{\rho + q_h}.$$

By Danskin's theorem, $\nabla_\lambda \mathcal{D}_\rho(x, \lambda) = \mathcal{A}(x) - y(x, \lambda)$.

We now prove the formula for λ^k by induction. It holds for $k = 0$ because $\lambda^0 = 0$. Assume that it holds for some $k \ge 0$. Then $\lambda^k \in [0, q_h a_0]$, so $y^{k+1} = y(x^{k+1}, \lambda^k) = y_{\lambda^k}$. Together with $\mu^k = 0$, the dual update in (3.4) reads

$$\lambda^{k+1} = \lambda^k + \tau\Big[\frac{q_h a_0 - \lambda^k}{\rho + q_h} - \frac{\lambda^k}{s}\Big] = \varphi\lambda^k + \frac{\tau q_h a_0}{\rho + q_h}.$$

A direct computation shows that $\bar{\lambda} = \varphi\bar{\lambda} + \tau q_h a_0/(\rho + q_h)$. Subtracting the two identities gives $\lambda^{k+1} - \bar{\lambda} = \varphi(\lambda^k - \bar{\lambda}) = -\bar{\lambda}\varphi^{k+1}$, which completes the induction.

(ii) Since $J_{\mathcal{A}} = 0$, (2.2) gives $R_k = \max\{\|\nabla f(x^{k+1})\|, \|\mathcal{A}(x^{k+1}) - y^{k+1}\|\}$. By (i) and $q_h a_0 - \bar{\lambda} = q_h a_0(\rho + q_h)/(\rho + q_h + s)$,

$$\mathcal{A}(x^{k+1}) - y^{k+1} = \frac{q_h a_0 - \lambda^k}{\rho + q_h} = \frac{q_h a_0}{\rho + q_h + s}\Big(1 + \frac{s}{\rho + q_h}\varphi^k\Big) = 4\epsilon\Big(1 + \frac{s}{\rho + q_h}\varphi^k\Big),$$

where the last equality uses $\rho + q_h + s = \rho + 3s + 1$. By (5.3), $\frac{s}{\rho+q_h}\varphi^k > -\frac{1}{4}$, so $\|\mathcal{A}(x^{k+1}) - y^{k+1}\| > 3\epsilon$. Thus no output is an ϵ-KKT point, and $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, 0; I) = +\infty$. ∎

5.5. Proof of Theorem 4.2. We now combine the three instances. Let c_1 and c_2 be given by (A.4). The constant c_1 depends only on (γ, η, τ), and c_2 depends only on (γ, τ, Δ), as required in Theorem 4.2.

Proof of Theorem 4.2. Fix $\epsilon \in (0, c_2)$, $\rho \geq 0$, and $s \geq \tau$ with $\epsilon(\rho + s) \leq c_1$, and fix $(\alpha, \beta) \in [0, 1]^2$. The definition of c_2 gives $\epsilon \leq \frac{1}{64} \leq \frac{1}{2}$, $\epsilon \leq \sqrt{\Delta}/2$, and $\epsilon^2 \leq \Delta/(64\gamma) \leq \Delta/(16\gamma)$. Hence the conditions of Example 5.1, Lemma 5.1, and Example 5.3 on ϵ hold. We treat the three parts of the partition of $[0, 1]^2$ separately.

Case 1: $\alpha \leq \alpha_\epsilon$. Let I be the instance of Example 5.1, which belongs to $\mathcal{I}(\mathbf{C}_0, \Delta)$. If $\alpha = 0$, Lemma 5.1 (iii) gives $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) = +\infty$. If $0 < \alpha \leq \alpha_\epsilon$, Lemma 5.1 (iii) and the definition of α_ϵ give

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) > \frac{\Delta}{16\gamma\alpha\epsilon^2} \geq \frac{\Delta}{16\gamma\alpha_\epsilon\epsilon^2} = \frac{\Delta\,\epsilon^{-4}}{16384\,\gamma\max\{1, 4\tau, 4\gamma\tau\}}.$$

Both bounds hold for every $\beta \in [0, 1]$.

Case 2: $\alpha > \alpha_\epsilon$ and $\beta > 0$. The conditions of Lemma 5.3 hold. Let I be the instance of Example 5.2. By Lemma 5.3 (i), $I \in \mathcal{I}(\mathbf{C}_0, \Delta)$, and Lemma 5.4 gives $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) = +\infty$.

Case 3: $\alpha > \alpha_\epsilon$ and $\beta = 0$. The conditions of Lemma 5.5 hold. Let I be the instance of Example 5.3, which belongs to $\mathcal{I}(\mathbf{C}_0, \Delta)$. Lemma 5.5 (ii) gives $T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, 0; I) = +\infty$.

The three cases cover $[0, 1]^2$. Hence, for every $(\alpha, \beta) \in [0, 1]^2$, some $I \in \mathcal{I}(\mathbf{C}_0, \Delta)$ satisfies

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) \geq \frac{\Delta\,\epsilon^{-4}}{16384\,\gamma\max\{1, 4\tau, 4\gamma\tau\}}.$$

Taking the supremum over I and then the infimum over (α, β) in Definition 3.3 gives the bound on $T_\epsilon^{\mathrm{KKT}}(\mathcal{C}_\theta, \mathcal{I}(\mathbf{C}_0, \Delta))$. ∎

In Cases 2 and 3, the method never produces an ϵ-KKT output. The finite bound comes only from Case 1, and it determines the order ϵ^{-4} in Theorem 4.2.

Appendix A. Details for the rotating instance.

A.1. Parameters. Let $V := \beta + \frac{\tau}{s}(1 - \beta)$ and

$$\nu := \frac{64\eta\tau\alpha\beta\epsilon^2}{\big(\alpha + \frac{\eta}{\gamma}(1 - \alpha)\big)V}, \qquad \zeta := 1 - \frac{\nu}{2} + i\sqrt{\nu - \frac{\nu^2}{4}}. \tag{A.1}$$

Under the conditions of Lemma 5.3, $0 < \nu < 4$; hence $\zeta \notin \mathbb{R}$, $|\zeta| = 1$, and $|\zeta - 1|^2 = \nu$. Let $W := 1 + \frac{\eta}{\gamma}\alpha(1 - \alpha)/|\zeta - 1 + \alpha|^2$, and define

$$b^2 := \frac{\nu W|\zeta - 1 + V|^2}{\eta\tau\big(\beta V + (1 - V)\nu\big)}, \qquad a := \frac{\nu(W - 1)}{\eta\alpha} + \frac{\tau b^2(V - \beta)}{|\zeta - 1 + V|^2}, \qquad \kappa := \rho b^2 + a. \tag{A.2}$$

Let $X > 0$ be such that $\max\{X, \|\Lambda\|\} = \frac{1}{3}$, where $\Lambda := \frac{\tau b\zeta(\zeta-1+\beta)}{(\zeta-1)(\zeta-1+V)}X$, and set

$$\boldsymbol{v} := \Big(X,\ \Lambda,\ \frac{\alpha\zeta}{\zeta-1+\alpha}X,\ \frac{\beta\zeta}{\zeta-1+\beta}\Lambda\Big). \tag{A.3}$$

The shifts x_c and λ_c are given in (A.8). They are chosen so that $-\boldsymbol{w}_c - \boldsymbol{v}$ lies in the $\mathbb{T}$-invariant subspace on which the spectral radius of $\mathbb{T}$ is less than one.

We use the following choices of the constants in Theorem 4.2:

$$\begin{aligned} c_1 &:= \min\left\{\frac{1}{32}\sqrt{\frac{\tau}{\eta(1+4\gamma/\eta)}},\ \frac{1}{64(\gamma/\eta+5/2)},\ \frac{1}{16(3+\tau^{-1})}\right\},\\ c_2 &:= \min\left\{\frac{1}{64},\ \frac{1}{32\sqrt{\max\{1,4\tau,4\gamma\tau\}}},\ \frac{\Delta}{12},\ \frac{\sqrt{\Delta}}{2},\ \sqrt{\frac{\Delta}{64\gamma}}\right\}. \end{aligned} \tag{A.4}$$

Throughout this appendix, assume the parameter conditions of Lemma 5.3. All complex scalars represent vectors in $\mathbb{R}^2$, with the Euclidean norm equal to the complex modulus. For the calculations below, write

$$\chi := \frac{\eta}{\gamma}, \qquad d_\tau := \frac{\tau}{s}, \qquad U := \alpha + \chi(1-\alpha), \qquad D := \alpha W,$$

$$E := |\zeta - 1 + \alpha|^2 = \alpha^2 + (1-\alpha)\nu, \qquad H := |\zeta - 1 + V|^2 = V^2 + (1-V)\nu,$$

$$\Upsilon := \beta V + (1-V)\nu, \qquad \xi := \frac{\eta\tau b^2(V-\beta)}{H}, \qquad C_{\mathrm{tr}} := \frac{1}{d_\tau} + \frac{2(1+d_\tau^{-1})}{\chi}.$$

These quantities are used only to prove the estimates and spectral identities.

A.2. Parameter bounds and proof of Lemma 5.3 (i).

LEMMA A.1. *Under the conditions of Lemma 5.3, the parameters satisfy*

$$7\epsilon \le b \le 8\epsilon, \qquad \max\{X, \|\Lambda\|\} = \frac{1}{3},$$

$$aX \le \frac{1}{8}b\|\Lambda\| + sb^2X,$$

$$bs\Big(\frac{\gamma}{\eta}+\frac{5}{2}\Big) \le \frac{1}{8}, \qquad b\Big(2s\Big(\frac{\gamma}{\eta}+\frac{5}{4}\Big)+\frac{5}{4}\rho\Big) \le \frac{1}{4}.$$

Proof of Lemma A.1. We first establish the bounds needed in all three proofs. Since $U \ge \chi$, $V \ge \beta$, and $\alpha > 1024\max\{1, 4\tau, 4\gamma\tau\}\epsilon^2$,

$$0 < \frac{\nu}{\alpha^2} \le \frac{64\gamma\tau\epsilon^2}{\alpha} < \frac{1}{64}.$$

In particular, $0 < \nu < 4$, so ζ is well defined and $|\zeta| = 1$. The definition of W gives

$$\frac{64}{65}U \le D = \alpha + \frac{\chi(1-\alpha)}{1+(1-\alpha)\nu/\alpha^2} \le U \le 1, \qquad D \ge \frac{\chi}{2}.$$

Since $V\Upsilon - \beta H = (1-V)(V-\beta)\nu \ge 0$,

$$\frac{b^2}{64\epsilon^2} = \frac{D}{U}\frac{\beta H}{V\Upsilon} \le 1.$$

Also, $H \ge V^2$, $V \ge d_\tau$, and $V - \beta \le V$ imply $0 \le \xi \le \eta s b^2$. Using $C_{\mathrm{tr}} \le (1 + 4/\chi)/d_\tau$ and (A.4), we obtain

$$\xi C_{\mathrm{tr}} \le \frac{64\eta(1+4/\chi)}{\tau}(\epsilon s)^2 \le \frac{1}{16}. \tag{A.5}$$

Thus $\xi/d_\tau \le 1/16$. The identity

$$\frac{\beta H}{V\Upsilon} = 1 - \frac{(1-V)\xi}{WV} \ge \frac{15}{16}$$

now yields $b^2/(64\epsilon^2) \ge 12/13 > 49/64$, proving $7\epsilon \le b \le 8\epsilon$. The normalization of X gives $\max\{X, \|\Lambda\|\} = 1/3$.

To bound a, set $a_{\mathrm{p}} := \nu(W-1)/(\eta\alpha)$ and $a_{\mathrm{d}} := \xi/\eta$. The definitions imply

$$\eta a = \frac{\nu(W-1)}{\alpha} + \xi, \qquad \eta\tau b^2 = \nu W + V\xi, \qquad U - D = \frac{(1-\alpha)\nu(W-1)}{\alpha}. \tag{A.6}$$

Hence $a = a_{\mathrm{p}} + a_{\mathrm{d}}$, $a_{\mathrm{p}} \le \tau b^2/\alpha$, and $a_{\mathrm{d}} \le s b^2$. Since $c_1 \le 1/224$ and $\epsilon < 1/64$,

$$0 < \kappa \le (\rho + s)b^2 + \frac{\tau b^2}{\alpha} \le 64c_1\epsilon + \frac{\tau}{16\max\{1, 4\tau, 4\gamma\tau\}} \le \frac{1}{32}.$$

For the amplitude bounds, put $E_\beta := \beta^2 + (1-\beta)\nu$. The formula for Λ gives $\|\Lambda\|^2/X^2 = \tau W E_\beta/(\eta\Upsilon)$. Moreover,

$$\Upsilon \le \mathrm{Re}\big((\zeta - 1 + V)\overline{(\zeta - 1 + \beta)}\big) \le \sqrt{HE_\beta},$$

where the difference in the first inequality is $(V-\beta)\nu/2$. Consequently,

$$\frac{a_{\mathrm{p}} X}{b\|\Lambda\|} = \frac{\sqrt{\nu}(W-1)\Upsilon}{\alpha W\sqrt{HE_\beta}} \le \frac{\sqrt{\nu}}{\alpha} \le \frac{1}{8},$$

$$\frac{a_{\mathrm{p}} X^2}{b^2\|\Lambda\|^2} = \frac{\eta(W-1)\Upsilon^2}{\alpha W^2 H E_\beta} \le \frac{\eta}{D} \le 2\gamma.$$

Adding the bound on a_{d} proves the asserted bound on aX and also gives

$$\kappa X^2 \le (\rho + s + 2\gamma)b^2 \max\{X^2, \|\Lambda\|^2\}. \tag{A.7}$$

Finally, $b \le 8\epsilon$ and (A.4) imply

$$bs\Big(\frac{1}{\chi} + \frac{5}{2}\Big) \le 8c_1\Big(\frac{1}{\chi} + \frac{5}{2}\Big) \le \frac{1}{8},$$

$$b\Big(2s\Big(\frac{1}{\chi} + \frac{5}{4}\Big) + \frac{5}{4}\rho\Big) \le 16c_1\Big(\frac{1}{\chi} + \frac{5}{4}\Big) \le \frac{1/\chi + 5/4}{4(1/\chi + 5/2)} < \frac{1}{4}.$$

∎

Proof of Lemma 5.3 (i). We use the bounds on $\mathbb{T}^k \boldsymbol{r}$ proved in Appendix A.4; that proof uses only Lemma A.1 and the spectral identities, and does not use membership in the problem class. At $k = 0$, those bounds and (A.8) give

$$\|x_c\| \le \frac{5}{4}X = R_*, \qquad \|\lambda_c\| \le \|\Lambda\| + s\Big(\frac{1}{\chi} + \frac{5}{4}\Big)bX \le \frac{1}{3}\left(1 + \frac{1/\chi + 5/4}{8(1/\chi + 5/2)}\right) < \frac{1}{2}.$$

The function ψ and its first derivative agree at R_* and $3R_*$. On each smooth radial piece, the Hessian of $\psi(\|x - x_c\|)$ has radial eigenvalue $\psi''(r)$ and tangential eigenvalue $\psi'(r)/r$, both of absolute value at most κ. Near x_c, this function is quadratic. Integrating its continuous gradient along line segments therefore shows that $\mathrm{Lip}(\nabla f) \le \kappa \le 1/32$. Also $|\psi'(r)| \le \kappa R_*$ for every $r \ge 0$, so

$$\|\nabla f(x)\| \le \kappa R_* + b\|\lambda_c\| \le \frac{5}{384} + \frac{1}{2} < 1.$$

Here we used $R_* \le 5/12$ and $b \le 8\epsilon < 1$. The affine mapping $\mathcal{A}$ has constant Jacobian of norm $b \le 1$, and $h = \|\cdot\|_2$ is convex and 1-Lipschitz.

It remains to bound the initial gap. Since $\psi(r) \ge -\kappa R_*^2$ and $\|\lambda_c\| < 1/2$,

$$F(x) = \psi(\|x - x_c\|) - b\langle \lambda_c, x - x_c\rangle + b\|x - x_c\| \ge -\kappa R_*^2.$$

At the origin, $\|x_c\| \le R_*$ and $\psi(\|x_c\|) \le 0$, whence

$$F(0) - \inf F \le \frac{3}{2} bR_* + \kappa R_*^2 \le 5\epsilon + \frac{100}{9}(\rho + s + 2\gamma)\epsilon^2.$$

By $\epsilon(\rho + s) \le c_1 \le 1/224$, this is at most $6\epsilon + 32\gamma\epsilon^2 \le \Delta$, where the last inequality uses $\epsilon < c_2$. Thus the instance belongs to $\mathcal{I}(\mathbf{C}_0, \Delta)$. ∎

A.3. Proof of Lemma 5.2. Define

$$\begin{aligned} r_x &:= -\frac{(W-1)\frac{\zeta-1}{\zeta-1+\alpha} + \frac{\xi\zeta}{\zeta-1+V}}{W+\xi} X, \\ r_\lambda &:= \frac{1}{\eta b}\Big[\frac{\eta}{\gamma}\frac{(1-\alpha)(\zeta-1)}{\zeta-1+\alpha} X + (\eta a + \alpha W) r_x - \xi\Big((1-\alpha) r_x + \frac{(V-\alpha)\zeta}{\zeta-1+V} X\Big)\Big]. \end{aligned}$$

Set

$$x_c := -(X + r_x), \qquad \lambda_c := -(\Lambda + r_\lambda), \qquad \boldsymbol{r} := -\boldsymbol{w}_c - \boldsymbol{v}. \tag{A.8}$$

Proof. For this calculation, write $Z := \boldsymbol{v}_z$, $M := \boldsymbol{v}_\mu$, and $\vartheta := (1 - V)\xi$. The characteristic polynomial factors as

$$\begin{aligned} \det((1+t)I - \mathbb{T}) &= t^2(t+U)(t+V) - \eta a\, t(t+\alpha)(t+V) \\ &\quad + \eta\tau b^2 (t+1)(t+\alpha)(t+\beta) \\ &= (t^2 + \nu t + \nu)\big((t+D)(t+V) - \vartheta(t+\alpha)\big). \end{aligned} \tag{A.9}$$

The first equality follows by eliminating the two center coordinates in (5.2); expansion using (A.6) gives the second equality. Since $(\zeta - 1)^2 + \nu(\zeta - 1) + \nu = 0$, substitution of (A.3) in the four rows of $\mathbb{T}$ gives

$$\mathbb{T}\boldsymbol{v} = \zeta \boldsymbol{v}.$$

For example, the center rows follow from $(\zeta - 1 + \alpha)Z = \alpha\zeta X$ and $(\zeta - 1 + \beta)M = \beta\zeta\Lambda$; the dual row reduces to $(\zeta - 1)(\zeta - 1 + V)\Lambda = \tau b\zeta(\zeta - 1 + \beta)X$. The primal row then follows from the first factor in (A.9). The bounds on ν proved above give $\zeta \notin \mathbb{R}$ and $|\zeta| = 1$.

We next prove

$$Q_{\mathrm{f}}(\mathbb{T})\boldsymbol{r} = 0, \qquad Q_{\mathrm{f}}(t) := (t-1+D)(t-1+V) - \vartheta(t-1+\alpha). \tag{A.10}$$

First consider distinct roots t_1, t_2 of Q_{f}, with $\alpha < 1$, $\beta < 1$, $d_\tau < 1$, $V \neq \alpha$, and all following denominators nonzero. An eigenvector associated with a root t is

$$\left(1,\ \frac{\tau b t(t-1+\beta)}{(t-1)(t-1+V)},\ \frac{\alpha t}{t-1+\alpha},\ \frac{\tau b \beta t^2}{(t-1)(t-1+V)}\right)^{\top}.$$

Let its coefficients at t_1, t_2 be c_1', c_2'. Requiring that their sum with $\boldsymbol{v}$ have equal first and third coordinates, and equal second and fourth coordinates, gives

$$\sum_{j=1}^{2} c_j' \frac{t_j - 1}{t_j - 1 + \alpha} = -\frac{\zeta - 1}{\zeta - 1 + \alpha} X, \qquad \sum_{j=1}^{2} c_j' \frac{t_j}{t_j - 1 + V} = -\frac{\zeta}{\zeta - 1 + V} X.$$

To solve these two equations, set $S_0 := c_1' + c_2'$, $S_1 := \sum_j (t_j - 1) c_j'$, $C_{\mathrm{f}} := D + V - \vartheta$, and $D_{\mathrm{f}} := DV - \alpha\vartheta$. At a root $v = t_j - 1$ of $v^2 + C_{\mathrm{f}} v + D_{\mathrm{f}}$,

$$\frac{v}{v+\alpha} = \frac{D_{\mathrm{f}} + \alpha v}{\alpha^2 - C_{\mathrm{f}}\alpha + D_{\mathrm{f}}}, \qquad \frac{v+1}{v+V} = \frac{D_{\mathrm{f}} - C_{\mathrm{f}} + V - (1-V)v}{V^2 - C_{\mathrm{f}} V + D_{\mathrm{f}}}.$$

The two denominators equal $\alpha(W-1)(V-\alpha)$ and $\vartheta(V-\alpha)$, respectively. The resulting system has coefficient determinant $\alpha(1-V)(\alpha - V)(W + \xi) \neq 0$, and its solution is

$$S_0 = r_x, \qquad S_1 = -D r_x + \xi \left((1-\alpha) r_x + \frac{(V-\alpha)\zeta}{\zeta - 1 + V} X \right).$$

The primal row of $\mathbb{T}$ now gives $\eta b r_\lambda = \chi(X - Z) + \eta a r_x - S_1$, which is the stated formula for r_λ. The center synchronization gives $r_z = X - Z + r_x$ and $r_\mu = \Lambda - M + r_\lambda$. Thus this linear combination is precisely $\boldsymbol{r}$, proving (A.10) in the stated nondegenerate case.

All entries of $Q_{\mathrm{f}}(\mathbb{T})\boldsymbol{r}$ are continuous functions of the admissible parameters: $\alpha, b, V > 0$, $W + \xi > 0$, and $|\zeta - 1 + \alpha|, |\zeta - 1 + V| > 0$. To justify the density used here, first take $\alpha, \beta, d_\tau \in (0,1)$ and $V \neq \alpha$. Then $W - 1 > 0$ and $\vartheta > 0$, so the two displayed denominators and the coefficient determinant are nonzero. Moreover,

$$Q_{\mathrm{f}}(1) = DV - \alpha\vartheta = \frac{\alpha\beta\eta\tau b^2}{\nu} > 0,$$

so neither root equals one. The remaining exclusion is a repeated root. For fixed α, β, d_τ, its discriminant

$$(D + V - \vartheta)^2 - 4(DV - \alpha\vartheta)$$

is a rational function of ϵ^2 and tends to $(U - V)^2$ as $\epsilon \downarrow 0$. If $U \neq V$, this rational function is not identically zero, so its zeros are isolated. Since $V = \beta + d_\tau(1 - \beta)$ varies strictly with β in the interior, the restrictions $V \neq \alpha$ and $V \neq U$ also define a dense subset. These perturbations can be made within the admissible parameter set: first decrease ϵ slightly if $\epsilon(\rho + s) = c_1$, and then perturb the other parameters. Taking limits therefore proves (A.10) also when roots coincide, $V = \alpha$, or $\alpha = 1$, $\beta = 1$, or $d_\tau = 1$.

To locate the roots of Q_{f}, note that $0 < D, V \le 1$ and, by (A.5),

$$\frac{2\vartheta}{DV} \le \frac{4\xi}{\chi d_\tau} \le 2\xi C_{\mathrm{tr}} \le \frac{1}{8}.$$

If $|t| \ge 1$, then

$$|(t-1+D)(t-1+V)| \ge DV|t|^2 > 2\vartheta|t| \ge |\vartheta(t-1+\alpha)|.$$

Thus every root of Q_{f} lies inside the unit circle. The subspace $\ker Q_{\mathrm{f}}(\mathbb{T})$ is invariant under $\mathbb{T}$, and every eigenvalue of the restriction is a root of Q_{f}. It contains $\boldsymbol{r}$ by (A.10), proving (ii), including repeated roots. Finally, $\boldsymbol{v} + \boldsymbol{r} = -\boldsymbol{w}_c$ follows from (A.8), proving (iii). ∎

A.4. Proof of Lemma 5.3 (ii).

Proof. Write $(\mathbb{T}^k\boldsymbol{r})_x$, $(\mathbb{T}^k\boldsymbol{r})_\lambda$, and $(\mathbb{T}^k\boldsymbol{r})_\mu$ as p_k, ℓ_k, m_k, respectively. These are components of the linear recursion, not new algorithm iterates. By the preceding spectral proof, all these components tend to zero. Put $r_{\mathrm{p}} := 1 - D$, $r_{\mathrm{d}} := 1 - V$, and $\vartheta := (1-V)\xi$. Equation (A.10) gives

$$p_{k+2} - (r_{\mathrm{p}} + r_{\mathrm{d}})p_{k+1} + r_{\mathrm{p}}r_{\mathrm{d}}p_k = \vartheta\big(p_{k+1} - (1-\alpha)p_k\big).$$

The initial values are $p_0 = r_x$ and $p_1 = r_{\mathrm{p}}r_x + \omega_x$, where

$$\omega_x := \xi\left((1-\alpha)r_x + \frac{(V-\alpha)\zeta}{\zeta - 1 + V}X\right).$$

Since $|\zeta - 1| = \sqrt{\nu}$, the formula for r_x implies

$$|r_x| \le \left(\frac{\sqrt{\nu}}{\alpha} + \frac{\xi}{d_\tau}\right)X \le \frac{3X}{16}.$$

As $|V - \alpha| \le 1$, it follows that $|\omega_x| \le \xi(1 + d_\tau^{-1})X$. Hence

$$|r_x| + \frac{|\omega_x|}{D} \le \left(\frac{1}{8} + \xi C_{\mathrm{tr}}\right)X \le \frac{3X}{16}. \tag{A.11}$$

For $j \ge 0$, define the convolution kernel $G_j := \sum_{i=0}^{j} r_{\mathrm{p}}^{j-i} r_{\mathrm{d}}^{i}$. Since $0 \le r_{\mathrm{p}}, r_{\mathrm{d}} < 1$,

$$0 \le G_j \le \sum_{i=0}^{j} r_{\mathrm{p}}^{j-i} \le \frac{1}{D}, \qquad \sum_{j\ge 0} G_j = \frac{1}{DV}.$$

Solving the scalar recurrence gives, for $k \ge 1$,

$$p_k = r_{\mathrm{p}}^k r_x + G_{k-1}\omega_x + \vartheta\sum_{j=0}^{k-2} G_{k-2-j}\big(p_{j+1} - (1-\alpha)p_j\big),$$

with an empty sum when $k = 1$. This formula follows by substitution in the recurrence and its two initial values. For each finite N, let $M_N := \max_{0\le k\le N}|p_k|$. Using (A.11) and $2\vartheta/(DV) \le 1/4$,

$$M_N \le \frac{3X}{16} + \frac{2\vartheta}{DV}M_N, \qquad M_N \le \frac{X}{4}.$$

Since N is arbitrary, this proves the primal bound. Summing absolute values on a finite prefix instead gives

$$\sum_{k=0}^{N} |p_k| \le \frac{|r_x|}{D} + \frac{|\omega_x|}{DV} + \frac{2\vartheta}{DV} \sum_{k=0}^{N} |p_k|.$$

Because $D, V \le 1$,

$$\frac{|r_x|}{D} + \frac{|\omega_x|}{DV} \le \frac{|r_x| + |\omega_x|/D}{DV}.$$

Letting N increase yields

$$\sum_{k\ge 0} |p_k| \le \frac{X}{4DV} \le \frac{X}{\chi d_\tau}. \tag{A.12}$$

For the dual bound, introduce the difference and weighted average

$$e_k := \ell_k - m_k, \qquad q_k := \frac{\beta(1-d_\tau)\ell_k + d_\tau m_k}{V}.$$

The dual and center rows of $\mathbb{T}$ give

$$e_{k+1} = (1-V)e_k + \tau b(1-\beta)p_{k+1}, \qquad q_{k+1} = q_k + \frac{\tau b\beta}{V} p_{k+1}.$$

The synchronization at $k = 0$ implies

$$e_0 = -\frac{\tau b(1-\beta)\zeta}{\zeta - 1 + V} X, \qquad |e_0| \le sbX.$$

Iterating the first recurrence and using $|p_k| \le X/4$ gives

$$|e_k| \le sbX + \frac{\tau b(1-\beta)}{V} \frac{X}{4} \le \frac{5}{4} sbX.$$

Since $q_k \to 0$, the second recurrence and the absolutely summable bound (A.12) give

$$q_k = -\frac{\tau b\beta}{V} \sum_{j=k+1}^{\infty} p_j, \qquad |q_k| \le \frac{s}{\chi} bX.$$

Finally, $\ell_k = q_k + (d_\tau/V)e_k$, with $d_\tau/V \le 1$. Thus

$$\|(\mathbb{T}^k \boldsymbol{r})_\lambda\| = |\ell_k| \le s\left(\frac{1}{\chi} + \frac{5}{4}\right) bX = s\left(\frac{\gamma}{\eta} + \frac{5}{4}\right) bX,$$

as required. ∎

Appendix B. Proof of Theorem 4.1.

B.1. Curvature and smoothness of the potential function. Let us define the constants

$$L_{\mathcal{D},\boldsymbol{x}} := L_f + L_{\mathcal{A}} L_h + \rho G_{\mathcal{A}}^2, \tag{B.1}$$

and

$$\mu_\gamma := \frac{1}{\gamma} - L_{\mathcal{D},\boldsymbol{x}}, \qquad L_x := L_{\mathcal{D},\boldsymbol{x}} + \frac{1}{\gamma}, \qquad L_\lambda(s) := \frac{1}{\rho} + \frac{1}{s},$$

where L_f, $L_{\mathcal{A}}$, $G_{\mathcal{A}}$, and L_h are the constants of Assumptions 2.1 and 2.2.

The following proposition summarizes the curvature and smoothness properties of the smoothed perturbed potential function $\mathcal{P}_s$. The proof is deferred to Appendix D.

PROPOSITION B.1. *Suppose that Assumptions 2.1 and 2.2 hold, and let the smoothing parameter* $0 < \gamma < \frac{1}{2L_{\mathcal{D},\boldsymbol{x}}}$. *Then, for all* $s > 0$, $\mathcal{P}_s$ *is* μ_γ*-strongly convex in* $\boldsymbol{x}$ *and* $1/s$*-strongly concave in* $\boldsymbol{\lambda}$. *Moreover, for all* $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^n$, $\boldsymbol{\lambda}, \boldsymbol{\lambda}' \in \mathbb{R}^m$, $\boldsymbol{z} \in \mathbb{R}^n$, *and* $\boldsymbol{\mu} \in \mathbb{R}^m$, *it holds that*

$$\big\|\nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}, \boldsymbol{\mu}) - \nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}', \boldsymbol{\lambda}', \boldsymbol{z}, \boldsymbol{\mu})\big\| \le L_x \|\boldsymbol{x} - \boldsymbol{x}'\| + G_{\mathcal{A}} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|, \tag{B.2}$$

$$\big\|\nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}, \boldsymbol{\mu}) - \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}', \boldsymbol{\lambda}', \boldsymbol{z}, \boldsymbol{\mu})\big\| \le G_{\mathcal{A}} \|\boldsymbol{x} - \boldsymbol{x}'\| + L_\lambda(s) \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|. \tag{B.3}$$

In particular, $\mathcal{P}_s$ *is* L_x*-smooth in* $\boldsymbol{x}$ *and* $L_\lambda(s)$*-smooth in* $\boldsymbol{\lambda}$.

B.2. Proof of the iteration complexity bound. Throughout this appendix, let

$$L := \sqrt{2}\, \max\Big\{L_{\mathcal{D},\boldsymbol{x}},\ G_{\mathcal{A}},\ \frac{1}{\rho}\Big\}, \qquad \bar{\gamma} := \frac{1}{4L}, \qquad \bar{\tau} := \frac{1}{36L}\Big(\frac{\eta}{\gamma}\Big)^2,$$

and let $\mathcal{Y} := \{\boldsymbol{\lambda} \in \mathbb{R}^m : \|\boldsymbol{\lambda}\| \le L_h + 1\}$. We write [ZXSL] for [37]; equation and lemma numbers refer to its arXiv version 2010.15768v3.

Step 1: reduction to Smoothed GDA. Since $\beta = 1$ and $\boldsymbol{\mu}^0 = \boldsymbol{\lambda}^0 = \mathbf{0}$, induction gives $\boldsymbol{\mu}^k = \boldsymbol{\lambda}^k$ for all k. Hence the gradient of the dual regularizer vanishes at every dual step, and the updates (3.4) read

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta\Big[\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) + \frac{1}{\gamma}(\boldsymbol{x}^k - \boldsymbol{z}^k)\Big], \qquad \boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \tau \nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k),$$

together with $\boldsymbol{z}^{k+1} = \boldsymbol{z}^k + \alpha(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k)$. None of these updates involves s. By (3.2), $\nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}) = (\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) - \boldsymbol{\lambda})/\rho$, so

$$\boldsymbol{\lambda}^{k+1} = \Big(1 - \frac{\tau}{\rho}\Big)\boldsymbol{\lambda}^k + \frac{\tau}{\rho}\, \boldsymbol{q}^k, \qquad \|\boldsymbol{q}^k\| \le L_h,$$

where the bound holds because $\boldsymbol{q}^k \in \partial h(\boldsymbol{y}^{k+1})$ and h is L_h-Lipschitz. Since $L \ge 1/\rho$, we have $\tau \le \bar{\tau} \le 1/(144L) \le \rho$, and induction gives $\|\boldsymbol{\lambda}^k\| \le L_h$ for all k. Hence every $\boldsymbol{\lambda}^k$ lies in the interior of $\mathcal{Y}$, and the projection onto $\mathcal{Y}$ does not change the iterates. Consequently, $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k)\}$ is the sequence generated by Smoothed GDA [37, Algorithm 2] applied to $\min_{\boldsymbol{x} \in \mathbb{R}^n} \max_{\boldsymbol{\lambda} \in \mathcal{Y}} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda})$ from zero, with smoothing parameter $p = 1/\gamma$, primal stepsize η, dual stepsize τ, and averaging parameter α.

Step 2: assumptions of [37, Theorem 3.4]. The proof of Proposition B.1 gives

$$\begin{aligned}
\|\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) - \nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}', \boldsymbol{\lambda}')\| &\le L_{\mathcal{D},\boldsymbol{x}} \|\boldsymbol{x} - \boldsymbol{x}'\| + G_{\mathcal{A}} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|, \\
\|\nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) - \nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}', \boldsymbol{\lambda}')\| &\le G_{\mathcal{A}} \|\boldsymbol{x} - \boldsymbol{x}'\| + \frac{1}{\rho} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|.
\end{aligned}$$

By the Cauchy–Schwarz inequality, both gradients are L-Lipschitz on $\mathbb{R}^n \times \mathbb{R}^m$. The set $\mathcal{Y}$ is convex and compact, $\mathbb{R}^n$ is closed and convex, and $\mathcal{D}_\rho(\boldsymbol{x}, \cdot)$ is concave. By Lemma 3.1,

$$\mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) = \max_{\boldsymbol{q}} \Big\{ \mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) - \frac{1}{2\rho}\|\boldsymbol{q} - \boldsymbol{\lambda}\|^2 \Big\} \le \max_{\boldsymbol{q}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) = F(\boldsymbol{x}),$$

with equality for $\boldsymbol{\lambda} \in \partial h(\mathcal{A}(\boldsymbol{x}))$. Every such $\boldsymbol{\lambda}$ satisfies $\|\boldsymbol{\lambda}\| \le L_h$ and hence lies in $\mathcal{Y}$. Therefore $\max_{\boldsymbol{\lambda} \in \mathcal{Y}} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) = F(\boldsymbol{x})$, which is bounded below by $\inf F > -\infty$. Thus Assumption 3.3 of [ZXSL] holds with gradient Lipschitz constant L.

Step 3: parameter conditions. We verify condition (3.3) of [ZXSL]. First, $\gamma \le \bar{\gamma}$ gives $p = 1/\gamma \ge 4L > 3L$. Second, since $\gamma L \le 1/4$, $\gamma/(1+\gamma L) \ge 4\gamma/5 > \gamma/2 \ge \eta$, hence $\eta < 1/(p+L)$. Third, put $a := \eta(p-L)$. Then $3\eta/(4\gamma) \le a < \eta p \le 1/2$, so

$$\frac{\eta^2 (p-L)^2}{4L(1+\eta(p-L))^2} > \frac{a^2}{16L} \ge \frac{9}{256L}\Big(\frac{\eta}{\gamma}\Big)^2 > \bar{\tau} \ge \tau,$$

and $\bar{\tau} \le 1/(144L) < 1/(11L)$. Finally, the averaging parameter $\alpha = c\epsilon^2 \le c$ satisfies $\alpha \le \min\{1/36, (p-L)^2/(384p(p+L)^2)\}$ by the choice of c in Step 5.

Step 4: initial gap. Let $K(\boldsymbol{x}, \boldsymbol{z}; \boldsymbol{\lambda}) := \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{z}\|^2$, $d(\boldsymbol{\lambda}, \boldsymbol{z}) := \min_{\boldsymbol{x}} K(\boldsymbol{x}, \boldsymbol{z}; \boldsymbol{\lambda})$, and $P(\boldsymbol{z}) := \min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda} \in \mathcal{Y}} K(\boldsymbol{x}, \boldsymbol{z}; \boldsymbol{\lambda})$. The potential function of [37, (4.1)] at the zero initialization is $\varphi^0 = K(\mathbf{0}, \mathbf{0}; \mathbf{0}) - 2d(\mathbf{0}, \mathbf{0}) + 2P(\mathbf{0})$. By Proposition B.1, $K(\cdot, \mathbf{0}; \mathbf{0})$ is μ_γ-strongly convex, and $\|\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\mathbf{0}, \mathbf{0})\| = \|\nabla f(\mathbf{0}) + J_{\mathcal{A}}(\mathbf{0})^\top \boldsymbol{q}(\mathbf{0}, \mathbf{0})\| \le G_f + G_{\mathcal{A}} L_h$. Hence

$$d(\mathbf{0}, \mathbf{0}) \ge \mathcal{D}_\rho(\mathbf{0}, \mathbf{0}) - \frac{(G_f + G_{\mathcal{A}} L_h)^2}{2\mu_\gamma}.$$

By Step 2, $P(\mathbf{0}) = \min_{\boldsymbol{x}}\{F(\boldsymbol{x}) + \frac{1}{2\gamma}\|\boldsymbol{x}\|^2\} \le F(\mathbf{0})$. Moreover, $\mathcal{D}_\rho(\mathbf{0}, \mathbf{0}) = f(\mathbf{0}) + \min_{\boldsymbol{y}}\{h(\boldsymbol{y}) + \frac{\rho}{2}\|\mathcal{A}(\mathbf{0}) - \boldsymbol{y}\|^2\}$, and the Lipschitz continuity of h gives $0 \le F(\mathbf{0}) - \mathcal{D}_\rho(\mathbf{0}, \mathbf{0}) \le L_h^2/(2\rho)$. Combining these bounds,

$$\varphi^0 - \inf F \le \hat{\Delta} := \Delta + \frac{L_h^2}{2\rho} + \frac{(G_f + G_{\mathcal{A}} L_h)^2}{\mu_\gamma},$$

where $\mu_\gamma \ge 1/\gamma - L \ge 3/(4\gamma) > 0$.

Step 5: rate. For $t \ge 0$, set

$$S_{t+1} := \max\big\{ \|\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^{t+1})\|,\ \|\nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^{t+1})\| \big\}.$$

Since $\boldsymbol{\lambda}^{t+1}$ lies in the interior of $\mathcal{Y}$, the normal cone of $\mathcal{Y}$ at $\boldsymbol{\lambda}^{t+1}$ is $\{\mathbf{0}\}$, and $(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^{t+1})$ is an S_{t+1}-solution in the sense of [37, Definition 3.1]. Fix $T \ge 1$. The proof of [37, Theorem 3.4] distinguishes the cases (B.52) and (B.53) and yields some $t \in \{0, \dots, T-1\}$ with either

$$\text{(i)} \quad S_{t+1} \le \lambda_1 \max\{\alpha, \sqrt{\alpha}\} \qquad \text{or} \qquad \text{(ii)} \quad S_{t+1} \le \lambda_2 \sqrt{\frac{\varphi^0 - \inf F}{T\alpha}},$$

by (B.62)–(B.63) with Lemma B.11 in case (i) and by Lemma B.12 in case (ii). Here λ_1 and λ_2 depend only on L, p, η, τ, and the diameter $2(L_h + 1)$ of $\mathcal{Y}$. Choose

$$c := \min\Big\{ \frac{1}{36},\ \frac{(p-L)^2}{384p(p+L)^2},\ \Big(\frac{10}{11\lambda_1}\Big)^2 \Big\}, \qquad T := \Big\lceil \Big(\frac{11}{10}\Big)^2 \frac{\lambda_2^2 \hat{\Delta}}{c\,\epsilon^4} \Big\rceil.$$

Since $\alpha = c\epsilon^2 \le 1$, case (i) gives $S_{t+1} \le \lambda_1 \sqrt{c}\,\epsilon \le \frac{10}{11}\epsilon$, and case (ii) gives $S_{t+1} \le \frac{10}{11}\epsilon$ by Step 4 and the choice of T.

Algorithm C.1 Smoothed-Perturbed Linearized Augmented Lagrangian Method (SPLALM)

Input: Initial $\boldsymbol{x}^0, \boldsymbol{z}^0 \in \mathbb{R}^n$ and $\boldsymbol{\lambda}^0, \boldsymbol{\mu}^0 \in \mathbb{R}^m$; Parameters: $\rho > 0$ and $\gamma > 0$; sequences: $\{\eta_k\}$, $\{\tau_k\}$, $\{\alpha_k\}$, and $\{\beta_k\}$; and iteration limit K.
for $k = 0, 1, \ldots, K-1$ **do**
  $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta_k \nabla_{\boldsymbol{x}} \mathcal{P}_{s_k}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k)$;
  $\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \tau_k \nabla_{\boldsymbol{\lambda}} \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k)$.
  $\boldsymbol{z}^{k+1} = \boldsymbol{z}^k + \alpha_k(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k)$.
  $\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \beta_k(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k)$.
end for

Step 6: from S_{t+1} to the KKT residual. By (2.2) and (3.2), $\mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{t+1}, \boldsymbol{y}^{t+1}, \boldsymbol{q}^t) = \max\{\|\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^t)\|, \|\mathbf{g}\|\}$ where $\mathbf{g} := \nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^t)$ and $\boldsymbol{\lambda}^{t+1} - \boldsymbol{\lambda}^t = \tau \mathbf{g}$. By Step 2 and $\tau/\rho \le 1/144$, $\|\nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^{t+1}) - \mathbf{g}\| \le \|\mathbf{g}\|/144$, hence $\|\mathbf{g}\| \le \frac{144}{143} S_{t+1}$. Similarly, $G_{\mathcal{A}}\tau \le 1/144$ gives $\|\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^{t+1}, \boldsymbol{\lambda}^t)\| \le S_{t+1} + G_{\mathcal{A}}\tau\|\mathbf{g}\| \le \frac{144}{143} S_{t+1}$. Therefore $\mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{t+1}, \boldsymbol{y}^{t+1}, \boldsymbol{q}^t) \le \frac{11}{10} S_{t+1} \le \epsilon$, and

$$T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, 1; I) \le t + 1 \le T \le C\epsilon^{-4}, \qquad C := \left(\frac{11}{10}\right)^2 \frac{\lambda_2^2 \hat{\Delta}}{c} + 1,$$

where the last inequality uses $\epsilon \le 1$. The constant c depends only on L, γ, η, τ, and L_h, hence only on θ and $\mathbf{C}$; the constant C depends in addition on Δ through $\hat{\Delta}$. Neither depends on the instance or on the dimensions. □

Appendix C. An accuracy-independent schedule with increasing s_k.

This appendix gives a parameter schedule that does not depend on ϵ. With ρ fixed, $s_k = \Theta((k+1)^{1/3})$, and diminishing center stepsizes, SPLALM finds an ϵ-KKT point in $\mathcal{O}(\epsilon^{-3})$ iterations (Theorem C.1). The parameters vary with k, so the schedule lies outside the family $\mathcal{C}_\theta$ of Definition 3.2.

Throughout this section, we additionally impose the following assumption. It is not part of the problem class $\mathcal{I}(\mathbf{C}, \Delta)$ of Definition 2.1.

Assumption C.1 (Level-bounded objective). For every $c \in \mathbb{R}$, the sublevel set $\{\boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \le c\}$ is empty or compact.

C.1. Algorithm with increasing s_k.

This section uses the updates of Definition 3.1 with iteration-dependent parameters s_k, η_k, τ_k, α_k, and β_k, and with an arbitrary initialization; see Algorithm C.1. By (3.1)–(3.3), the first two updates read

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta_k \Big[\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \boldsymbol{q}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) + \frac{1}{\gamma}(\boldsymbol{x}^k - \boldsymbol{z}^k)\Big], \\ \boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \tau_k \Big[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k) - \frac{1}{s_k}(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k)\Big], \end{aligned} \tag{C.1}$$

and the outputs $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ are those of Definition 3.1.

We use fixed primal and dual stepsizes together with diminishing center stepsizes. Choose $\rho > 0$ and $\gamma > 0$ satisfying Proposition B.1. With L_x and μ_γ defined in Section B.1 and $L_{d,0}$ defined in Section C.2, set $\eta_k \equiv \eta$ and $\tau_k \equiv \tau$. An admissible choice is

$$\begin{aligned} s_k &= c_s(k+1)^{1/3}, \qquad \eta = \frac{1}{2L_x}, \qquad \tau = \left(2L_{d,0} + \frac{72 L_x G_{\mathcal{A}}^2}{\mu_\gamma^2}\right)^{-1}, \\ \alpha_k &= \frac{\min\{s_0, \gamma\mu_\gamma\tau\}}{36 s_k}, \qquad \beta_k = \frac{\theta_\mu}{k + k_0}, \end{aligned} \tag{C.2}$$

where $c_s > 0$, $\theta_\mu \in (0,1]$, and $k_0 \geq 1$ are constants independent of K. Consequently,

$$s_k = \Theta((k+1)^{1/3}), \qquad \alpha_k = \Theta((k+1)^{-1/3}), \qquad \beta_k = \Theta((k+1)^{-1}).$$

THEOREM C.1. *Suppose that Assumptions C.1, 2.1, and 2.2 hold. Let the sequence generated by Algorithm C.1 use (C.2), with ρ and γ as specified above. For $K \geq 2$, choose a tail-best index*

$$\kappa \in \operatorname*{argmin}_{k \in \{\lceil K/2 \rceil, \ldots, K-1\}} \left\{ \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \alpha_k \|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \right\}.$$

Then it holds that

$$\max\left\{ \left[\left\| \nabla f(\boldsymbol{x}^{\kappa+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa+1})^\top \boldsymbol{q}^\kappa \right\|^2 \right], \left[\left\| \mathcal{A}(\boldsymbol{x}^{\kappa+1}) - \boldsymbol{y}^{\kappa+1} \right\|^2 \right] \right\} = \mathcal{O}(K^{-2/3}),$$

with $\boldsymbol{q}^\kappa \in \partial h(\boldsymbol{y}^{\kappa+1})$. Consequently, given $\epsilon > 0$, the iteration complexity of Algorithm C.1 to obtain an ϵ-KKT point is of order $\mathcal{O}(\epsilon^{-3})$.

C.2. Complexity Analysis. We use the constants $L_{\mathcal{D},\boldsymbol{x}}$, μ_γ, L_x, and $L_\lambda(s)$ and Proposition B.1 from Section B.1.

C.2.1. Value Functions and Lyapunov Function. Fix $s > 0$ and $(\boldsymbol{z}, \boldsymbol{\mu})$. By Proposition B.1, $\mathcal{P}_s(\cdot, \cdot; \boldsymbol{z}, \boldsymbol{\mu})$ is strongly convex in $\boldsymbol{x}$ and strongly concave in $\boldsymbol{\lambda}$. Hence the following value functions are well defined, and the minimizer and maximizer below are unique:

$$d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \min_{\boldsymbol{x} \in \mathbb{R}^n} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \qquad \boldsymbol{x}_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \operatorname*{argmin}_{\boldsymbol{x} \in \mathbb{R}^n} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \tag{C.3}$$

$$p_s(\boldsymbol{z}, \boldsymbol{\mu}) := \max_{\boldsymbol{\lambda} \in \mathbb{R}^m} d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \qquad \boldsymbol{\lambda}_s^\star(\boldsymbol{z}, \boldsymbol{\mu}) := \operatorname*{argmax}_{\boldsymbol{\lambda} \in \mathbb{R}^m} d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}). \tag{C.4}$$

Strong convexity–strong concavity also yields strong duality,

$$p_s(\boldsymbol{z}, \boldsymbol{\mu}) = \min_{\boldsymbol{x} \in \mathbb{R}^n} \max_{\boldsymbol{\lambda} \in \mathbb{R}^m} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \tag{C.5}$$

and the unique saddle point is $(\boldsymbol{x}_s^\star(\boldsymbol{z}, \boldsymbol{\mu}), \boldsymbol{\lambda}_s^\star(\boldsymbol{z}, \boldsymbol{\mu}))$, where

$$\boldsymbol{x}_s^\star(\boldsymbol{z}, \boldsymbol{\mu}) := \boldsymbol{x}_s\big(\boldsymbol{\lambda}_s^\star(\boldsymbol{z}, \boldsymbol{\mu}); \boldsymbol{z}, \boldsymbol{\mu}\big).$$

Following the three-value Lyapunov construction for smoothed nonconvex–concave minimax methods [37], and its perturbed-smoothed extension [14], we adapt the construction to the reduced augmented-Lagrangian setting and define

$$\begin{aligned} \Psi_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) &:= \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) - 2d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) + 2p_s(\boldsymbol{z}, \boldsymbol{\mu}) \\ &= \underbrace{\mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) - d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu})}_{\text{primal gap at fixed } \boldsymbol{\lambda}} + \underbrace{p_s(\boldsymbol{z}, \boldsymbol{\mu}) - d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu})}_{\text{dual gap}} + p_s(\boldsymbol{z}, \boldsymbol{\mu}). \end{aligned} \tag{C.6}$$

Both gaps are nonnegative, so $\Psi_s \geq p_s$.

Iteration-k shorthand.. At iteration k we evaluate the above objects at $(s, \boldsymbol{\mu}) = (s_k, \boldsymbol{\mu}^k)$, replace the subscript s by k, and suppress $\boldsymbol{\mu}^k$. For example, $\mathcal{P}_k(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}) := \mathcal{P}_{s_k}(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}^k)$ and $d_k(\boldsymbol{\lambda}; \boldsymbol{z}) := d_{s_k}(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}^k)$; the quantities p_k, $\boldsymbol{x}_k$, $\boldsymbol{\lambda}_k^\star$, $\boldsymbol{x}_k^\star$, and Ψ_k are defined analogously. We also write $\Delta_{\boldsymbol{x}}^{k+1} := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$, and define $\Delta_{\boldsymbol{\lambda}}^{k+1}$, $\Delta_{\boldsymbol{z}}^{k+1}$, and $\Delta_{\boldsymbol{\mu}}^{k+1}$ in the same way.

C.2.2. Basic One-Step Estimate. Throughout this subsection, we assume that

$$s_{k+1} \geq s_k > 0, \qquad 0 < \tau_k \leq s_k, \qquad 0 < \beta_k \leq 1, \qquad 0 \leq \alpha_k \leq 2. \tag{C.7}$$

The following lemma gives a basic one-step descent estimate of the potential function $\mathcal{P}$ from iteration k to $k+1$.

LEMMA C.1. *Suppose Assumptions 2.1 and 2.2 hold. Let the iterates be generated by Algorithm C.1 with the parameters satisfying (C.7). Then, for every $k \geq 0$, the following holds:*

$$\begin{aligned} &\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - \mathcal{P}_k(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) \\ \leq\ & -\left(\frac{1}{\eta_k} - \frac{L_x}{2}\right)\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \left(\frac{1}{\tau_k} - \frac{1}{2s_k}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ & - \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 + \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2. \end{aligned} \tag{C.8}$$

As shown in Lemma C.1, the positive ascent term generated by the $\boldsymbol{\lambda}$-update prevents the potential $\mathcal{P}_k$ from yielding a direct descent estimate. We therefore relate this term to the one-step increase of the dual value function, as developed in Lemma C.3. This relation allows us to construct a Lyapunov function with a descent property. Before proceeding, we establish several regularity properties of the value functions d_k and p_k, with their gradient formulas following from Danskin's theorem.

LEMMA C.2. *Under the conditions of Proposition B.1, for all $k \geq 0$, the following properties hold:*

(i) For every $\boldsymbol{z} \in \mathbb{R}^n$, the minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})$ is unique and $\frac{G_{\mathcal{A}}}{\mu_\gamma}$-Lipschitz continuous in $\boldsymbol{\lambda}$.

(ii) for every $\boldsymbol{z} \in \mathbb{R}^n$, $d_k(\cdot; \boldsymbol{z})$ is $1/s_k$-strongly concave and $L_{d,k}$-smooth, with gradient

$$\nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}; \boldsymbol{z}) = \nabla_{\boldsymbol{\lambda}} \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}), \boldsymbol{\lambda}; \boldsymbol{z}\big),$$

where $L_{d,k} := \frac{1}{\rho} + \frac{1}{s_k} + \frac{G_{\mathcal{A}}^2}{\mu_\gamma}$.

(iii) The function $p_k(z)$ is continuously differentiable, with gradient

$$\nabla p_k(\boldsymbol{z}) = \frac{1}{\gamma}\big(\boldsymbol{z} - \boldsymbol{x}_k^\star(\boldsymbol{z})\big),$$

and has a $1/\gamma$-Lipschitz continuous gradient.

The following lemma controls the dual ascent term.

LEMMA C.3. *Suppose that the conditions of Lemma C.2 hold, and let the iterates be generated by Algorithm C.1 with parameters satisfying (C.7). For every $k \geq 0$, assume that*

$$\frac{1}{\tau_k} + \frac{1}{2s_k} \geq 2L_{d,k}. \tag{C.9}$$

Then the following holds

$$\begin{aligned} &\left(\frac{1}{\tau_k} - \frac{1}{2s_k}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ &\leq 2\big[d_k(\boldsymbol{\lambda}^{k+1}; \mathbf{z}^k) - d_k(\boldsymbol{\lambda}^k; \mathbf{z}^k)\big] - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{4G_{\mathcal{A}}^2}{L_{d,k}\mu_\gamma}\left[\mathcal{P}_k(\mathbf{x}^{k+1}, \boldsymbol{\lambda}^k; \mathbf{z}^k) - d_k(\boldsymbol{\lambda}^k; \mathbf{z}^k)\right]. \end{aligned}$$

Moreover, it holds that

$$\begin{aligned}
&\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})-\mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\\
&\le -\left[\frac{1}{\eta_k}-\frac{L_x}{2}-\frac{2G_{\mathcal{A}}^2(1/\eta_k+L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2-\frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2\\
&\quad+2\big[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k)-d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\big]-\frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2\\
&\quad+\frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^k\|^2+\frac{1}{2}\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2.
\end{aligned}\tag{C.10}$$

The potential $\mathcal{P}_k$ increases along the dual update, so its variation alone does not capture progress in both variables. We therefore seek a merit function that accounts for both primal suboptimality and progress in dual maximization. To account for these two effects, consider the partial potential

$$\mathcal{P}_k-cd_k=(\mathcal{P}_k-d_k)-(c-1)d_k,\qquad c>1.\tag{C.11}$$

The first term measures the primal optimality gap, while the second term decreases as the dual value increases. In the following analysis, we set $c=2$, consistent with the coefficient of the dual-value increment in (C.10).

LEMMA C.4. *Under the conditions of Lemma C.3, for every $k\ge 0$, the following holds*

$$\begin{aligned}
&\Big[\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})-2d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})\Big]-\Big[\mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k)-2d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\Big]\\
&\le -\left[\frac{1}{\eta_k}-\frac{L_x}{2}-\frac{2G_{\mathcal{A}}^2(1/\eta_k+L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2-\frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2\\
&\quad+\frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2-\frac{2\alpha_k}{\gamma}\left\langle\boldsymbol{x}^{k+1}-\boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}),\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\right\rangle\\
&\quad+\frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^k\|^2+\frac{1}{2}\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2.
\end{aligned}\tag{C.12}$$

Equation (C.12) gives the one-step estimate for the partial potential in (C.11) with $c=2$. The $\boldsymbol{x}$- and $\boldsymbol{\lambda}$-updates contribute the first two terms on the right-hand side. The $\boldsymbol{z}$-update, however, produces a positive quadratic term and an indefinite inner-product term. Hence, the partial potential does not yet provide a sufficient descent estimate for the full iteration.

At an exact saddle point of $\min_{\boldsymbol{x}}\max_{\boldsymbol{\lambda}}\mathcal{P}_k(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z})$, strong duality gives

$$\mathcal{P}_k\big(\boldsymbol{x}_k^\star(\boldsymbol{z}),\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big)=d_k\big(\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big)=p_k(\boldsymbol{z}),$$

and therefore

$$\mathcal{P}_k\big(\boldsymbol{x}_k^\star(\boldsymbol{z}),\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big)-2d_k\big(\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big)=-p_k(\boldsymbol{z}).$$

Consequently, any $\boldsymbol{z}$-update that decreases the saddle value p_k increases the partial potential $\mathcal{P}_k-2d_k$. This motivates us to examine whether an additional value-function term can restore a descent estimate.

We therefore add the value-function increment to the partial potential in (C.11): $2[p_k(\boldsymbol{z}^{k+1})-p_k(\boldsymbol{z}^k)]$ and combine the two estimates directly. The resulting one-step bound is stated next.

LEMMA C.5. *Under the conditions of Lemma C.2 and $0\le\alpha_k\le 2$, the $\boldsymbol{z}$-update satisfies*

$$p_k(\boldsymbol{z}^{k+1})-p_k(\boldsymbol{z}^k)\le-\frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2+\frac{\alpha_k}{\gamma}\left\langle\boldsymbol{x}^{k+1}-\boldsymbol{x}_k^\star(\boldsymbol{z}^k),\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\right\rangle.\tag{C.13}$$

LEMMA C.6. *Under the conditions of Lemma C.4, for every $k \geq 0$,*

$$\begin{aligned}
&\Big[\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - 2d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) + 2p_k(\boldsymbol{z}^{k+1})\Big] \\
&\quad - \Big[\mathcal{P}_k(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - 2d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) + 2p_k(\boldsymbol{z}^k)\Big] \\
&\leq -\left[\frac{1}{\eta_k} - \frac{L_x}{2} - \frac{2G_{\mathcal{A}}^2(1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\
&\quad - \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 + \frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle \\
&\quad + \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2.
\end{aligned} \tag{C.14}$$

Two gaps remain before Φ_k admits a genuine descent inequality: the augmented estimate still carries an indefinite cross term and positive multiplier/center-increment terms, and its left-hand side compares d_k, p_k at both endpoints rather than the true $k \to k+1$ potential difference, which requires d_{k+1}, p_{k+1}.

For that, we establish boundedness of the multipliers directly, rather than assuming compactness a priori: both multipliers are controlled by the AL structure and the inner residual, not by projection onto a compact set.

LEMMA C.7. *Suppose Assumptions 2.1 and 2.2 hold. Let the iterates be generated by Algorithm C.1 with (C.7) and*

$$\tau_k\left(\frac{1}{\rho} + \frac{1}{s_k}\right) \leq 1. \tag{C.15}$$

Define the constant independent of k

$$M := \max\{\|\boldsymbol{\lambda}^0\|, \|\boldsymbol{\mu}^0\|, L_h\}.$$

Then, for every $k \geq 0$, the following holds

$$\|\boldsymbol{\lambda}^k\| \leq M, \qquad \|\boldsymbol{\mu}^k\| \leq M.$$

Moreover, for every $k \geq 0$ and every $\boldsymbol{z} \in \mathbb{R}^n$, it holds that

$$\|\boldsymbol{\lambda}_k^\star(\boldsymbol{z})\| \leq M.$$

LEMMA C.8. *Under the conditions of Proposition B.1, for every $k \geq 0$, $\boldsymbol{\lambda} \in \mathbb{R}^m$, and $\boldsymbol{z} \in \mathbb{R}^n$, it holds that*

$$\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}) - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2 \leq \frac{2}{\mu_\gamma}\big[p_k(\boldsymbol{z}) - d_k(\boldsymbol{\lambda}; \boldsymbol{z})\big]. \tag{C.16}$$

For any $\boldsymbol{z}, \boldsymbol{z}' \in \mathbb{R}^n$,

$$\max\big\{\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}') - \boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})\|, \|\boldsymbol{x}_k^\star(\boldsymbol{z}') - \boldsymbol{x}_k^\star(\boldsymbol{z})\|\big\} \leq \frac{1}{\gamma\mu_\gamma}\|\boldsymbol{z}' - \boldsymbol{z}\|. \tag{C.17}$$

We now establish an approximate descent inequality for Algorithm C.1 with respect to the Lyapunov function defined in (C.6). We assume the parameters are chosen as

$$\begin{aligned} &0<\gamma<\frac{1}{2L_{\mathcal{D},\boldsymbol{x}}}, \qquad 0<s_k\le s_{k+1}, \qquad 0<\beta_k\le 1, \qquad 0<\eta_k\le\frac{1}{2L_x},\\ &0<\tau_k\le\left(\max\left\{2L_{d,k},\,\frac{36G_{\mathcal{A}}^2}{\eta_k\mu_\gamma^2}\right\}\right)^{-1}, \qquad 0<\alpha_k\le\frac{1}{36}\min\left\{1,\,\frac{\gamma\mu_\gamma\tau_k}{s_k}\right\}, \end{aligned} \tag{C.18}$$

where $L_{d,k}:=\rho^{-1}+s_k^{-1}+G_{\mathcal{A}}^2/\mu_\gamma$. Since $L_{d,k}\ge\rho^{-1}+s_k^{-1}$, (C.18) implies the conditions of Proposition B.1, Lemma C.7, (C.7), (C.9), and (C.15).

LEMMA C.9. *Suppose that Assumptions C.1, 2.1, and 2.2 hold, and let* $\{(\boldsymbol{x}^k,\boldsymbol{\lambda}^k,\boldsymbol{z}^k,\boldsymbol{\mu}^k)\}$ *be generated by Algorithm C.1 with parameters satisfying* (C.18). *Then for every* $k\ge 0$*:*

(a) for all $\boldsymbol{x}\in\mathbb{R}^n$, $\boldsymbol{\lambda}\in\mathbb{R}^m$, *and* $\boldsymbol{z}\in\mathbb{R}^n$,

$$\Psi_k(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z})\ge p_k(\boldsymbol{z})\ge\inf_{\boldsymbol{x}\in\mathbb{R}^n}F(\boldsymbol{x})-\frac{(L_h+M)^2}{2s_0}; \tag{C.19}$$

(b) the iterates satisfy

$$\begin{aligned} &\Psi_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})-\Psi_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\\ &\quad\le-\frac{1}{4\eta_k}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2-\frac{1}{8\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2-\frac{\alpha_k}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2+\delta_k, \end{aligned} \tag{C.20}$$

where $\delta_k:=10M^2\big(s_k^{-1}-s_{k+1}^{-1}\big)+20M^2\beta_k/s_k$.

The proof is given in Appendix D.

Appendix D. Proofs for Appendix C.

Proof of the curvature part of Proposition B.1. For fixed $\boldsymbol{\lambda}$, the reduced payoff satisfies

$$\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})=\nabla f(\boldsymbol{x})+J_{\mathcal{A}}(\boldsymbol{x})^\top\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}),$$

where

$$\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})=\operatorname{prox}_{\rho h^*}\big(\boldsymbol{\lambda}+\rho\mathcal{A}(\boldsymbol{x})\big).$$

The nonexpansiveness of the proximal mapping, the bounded Jacobian, and the Lipschitz Jacobian imply

$$\|\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})-\boldsymbol{q}(\boldsymbol{x}',\boldsymbol{\lambda})\|\le\rho G_{\mathcal{A}}\|\boldsymbol{x}-\boldsymbol{x}'\|.$$

Assumption 2.2 gives $\|\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})\|\le L_h$. Therefore

$$\|\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})-\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x}',\boldsymbol{\lambda})\|\le L_{\mathcal{D},\boldsymbol{x}}\|\boldsymbol{x}-\boldsymbol{x}'\|.$$

Thus $\mathcal{D}_\rho(\cdot,\boldsymbol{\lambda})$ is $L_{\mathcal{D},\boldsymbol{x}}$-weakly convex. Adding $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$ gives μ_γ-strong convexity in $\boldsymbol{x}$.

For fixed $(\boldsymbol{x},\boldsymbol{z},\boldsymbol{\mu})$, $\mathcal{D}_\rho(\boldsymbol{x},\cdot)$ is concave in $\boldsymbol{\lambda}$ because it is the pointwise infimum of affine functions of $\boldsymbol{\lambda}$. The term $-\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ is $1/s$-strongly concave. Hence $\mathcal{P}_s$ is $1/s$-strongly concave in $\boldsymbol{\lambda}$. ∎

Proof of the smoothness part of Proposition B.1. The $\boldsymbol{x}$-block bound follows from the Lipschitz estimate for $\nabla_{\boldsymbol{x}}\mathcal{D}_\rho$ in the proof of Proposition B.1 and the quadratic term $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$.

For the $\boldsymbol{\lambda}$-block, the identity

$$\nabla_{\boldsymbol{\lambda}}\mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) = \frac{1}{\rho}\big(\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})-\boldsymbol{\lambda}\big) - \frac{1}{s}(\boldsymbol{\lambda}-\boldsymbol{\mu})$$

and the nonexpansiveness of $\mathrm{prox}_{\rho h^*}$ give $L_\lambda(s) = 1/\rho + 1/s$. The same estimate, together with

$$\|\mathcal{A}(\boldsymbol{x}) - \mathcal{A}(\boldsymbol{x}')\| \le G_{\mathcal{A}}\|\boldsymbol{x}-\boldsymbol{x}'\|,$$

gives both cross Lipschitz bounds with constant $G_{\mathcal{A}}$. ∎

Proof of Lemma C.3. Set

$$g^k := \nabla_{\boldsymbol{\lambda}}\mathcal{P}_k(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k,\boldsymbol{z}^k), \qquad \Delta_{\boldsymbol{\lambda}}^{k+1} = \tau_k g^k,$$

where the second identity follows from the dual update. By the $L_{d,k}$-smoothness of $d_k(\cdot;\boldsymbol{z}^k)$,

$$\begin{aligned} &d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k) \\ &\quad\ge \big\langle \nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k), \Delta_{\boldsymbol{\lambda}}^{k+1}\big\rangle - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2. \end{aligned}$$

Danskin's theorem and Lemma C.2 give

$$\nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k) = \nabla_{\boldsymbol{\lambda}}\mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k),\boldsymbol{\lambda}^k,\boldsymbol{z}^k\big).$$

The cross-Lipschitz estimate in Proposition B.1 therefore yields

$$\begin{aligned} &\big\langle \nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k), \Delta_{\boldsymbol{\lambda}}^{k+1}\big\rangle \\ &\quad= \tau_k \big\langle \nabla_{\boldsymbol{\lambda}}\mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k),\boldsymbol{\lambda}^k,\boldsymbol{z}^k\big), g^k\big\rangle \\ &\quad\ge \tau_k\|g^k\|^2 - \tau_k G_{\mathcal{A}}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\|\|g^k\|. \end{aligned}$$

Using

$$\tau_k\|g^k\|^2 = \frac{1}{\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2, \qquad \tau_k\|g^k\| = \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|,$$

we obtain

$$\begin{aligned} &d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k) \\ &\quad\ge \left(\frac{1}{\tau_k} - \frac{L_{d,k}}{2}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 - G_{\mathcal{A}}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\|\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|. \end{aligned} \tag{D.1}$$

Let

$$b_k := \frac{1}{\tau_k} + \frac{1}{2s_k} - L_{d,k}.$$

Multiplying (D.1) by 2 and rearranging gives

$$\begin{aligned} &\left(\frac{1}{\tau_k} - \frac{1}{2s_k}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ &\quad\le 2\big[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\big] - b_k\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ &\qquad + 2G_{\mathcal{A}}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\|\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|. \end{aligned}$$

The assumed parameter condition implies $b_k \geq L_{d,k} > 0$. Young's inequality with parameter $L_{d,k}$ gives

$$2G_{\mathcal{A}}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|\|\Delta_{\boldsymbol{\lambda}}^{k+1}\| \leq \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{2G_{\mathcal{A}}^2}{L_{d,k}}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2.$$

Since $b_k \geq L_{d,k}$, we have

$$-b_k\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \leq -\frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2.$$

Finally, the μ_γ-strong convexity of $\mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ gives

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2 \leq \frac{2}{\mu_\gamma}\left[\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\right].$$

Substituting this estimate proves the first inequality. ∎

Proof of the second estimate in Lemma C.3. Fix $k \geq 0$. By Proposition B.1, $\mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ is μ_γ-strongly convex. Applying the strong-convexity inequality at $\boldsymbol{x}^{k+1}$ to the minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ gives

$$\begin{aligned}
&\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \\
&\quad\leq \left\langle \nabla_{\boldsymbol{x}}\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\right\rangle \\
&\qquad - \frac{\mu_\gamma}{2}\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2 \\
&\quad\leq \frac{1}{2\mu_\gamma}\|\nabla_{\boldsymbol{x}}\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2,
\end{aligned}$$

where the last inequality follows from Young's inequality. The $\boldsymbol{x}$-update and the L_x-Lipschitz continuity of $\nabla_{\boldsymbol{x}}\mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ imply

$$\begin{aligned}
&\|\nabla_{\boldsymbol{x}}\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \\
&\quad\leq \|\nabla_{\boldsymbol{x}}\mathcal{P}_k(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| + L_x\|\Delta_{\boldsymbol{x}}^{k+1}\| \\
&\quad= \left(\frac{1}{\eta_k} + L_x\right)\|\Delta_{\boldsymbol{x}}^{k+1}\|.
\end{aligned}$$

Combining these estimates yields

$$\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \leq \frac{(1/\eta_k + L_x)^2}{2\mu_\gamma}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2. \tag{D.2}$$

Substituting (D.2) into the first inequality of Lemma C.3 gives a bound on the dual ascent term, where the coefficient of $\|\Delta_{\boldsymbol{x}}^{k+1}\|^2$ becomes

$$\frac{4G_{\mathcal{A}}^2}{L_{d,k}\mu_\gamma}\frac{(1/\eta_k + L_x)^2}{2\mu_\gamma} = \frac{2G_{\mathcal{A}}^2(1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}.$$

Substitute the resulting bound on $(1/\tau_k - 1/(2s_k))\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2$ into (C.8) and combine the coefficients of $\|\Delta_{\boldsymbol{x}}^{k+1}\|^2$. The $\boldsymbol{z}$-step term and both parameter-drift terms remain unchanged. This proves (C.10). ∎

Proof of Lemma C.4. Subtract $2[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)]$ from both sides of (C.10). To replace $d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k)$ on the left by $d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})$, add $-2[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k)]$ to both sides. By the minimum-value definition,

$$d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) \le \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}),\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k\big).$$

Subtracting this inequality from the corresponding equality at $\boldsymbol{z}^{k+1}$ and expanding the quadratic terms yields

$$\begin{aligned} &d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) \\ &\quad\ge \frac{1}{\gamma}\left\langle \boldsymbol{z}^k - \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}), \Delta_{\boldsymbol{z}}^{k+1}\right\rangle + \frac{1}{2\gamma}\|\Delta_{\boldsymbol{z}}^{k+1}\|^2 \\ &\quad= -\frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2 \\ &\qquad+ \frac{\alpha_k}{\gamma}\left\langle \boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}), \boldsymbol{x}^{k+1}-\boldsymbol{z}^k\right\rangle. \end{aligned}$$

Multiply this inequality by -2 and substitute it into the preceding estimate. Its squared-step term combines with the $\boldsymbol{z}$-step term in (C.10) to give $\alpha_k(2-\alpha_k)\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2/(2\gamma)$. Both parameter-drift terms remain unchanged, proving (C.12). ∎

Proof of Lemma C.5. By Lemma C.2, p_k has a $1/\gamma$-Lipschitz continuous gradient and

$$\nabla p_k(\boldsymbol{z}^k) = \frac{1}{\gamma}\big(\boldsymbol{z}^k - \boldsymbol{x}_k^\star(\boldsymbol{z}^k)\big).$$

The $\boldsymbol{z}$-update gives

$$\boldsymbol{z}^{k+1} - \boldsymbol{z}^k = \alpha_k(\boldsymbol{x}^{k+1}-\boldsymbol{z}^k).$$

Applying the smoothness inequality for p_k and expanding the inner product yields (C.13). ∎

Proof of Lemma C.6. By Lemma C.5,

$$\begin{aligned} 2\big[p_k(\boldsymbol{z}^{k+1}) - p_k(\boldsymbol{z}^k)\big] \le\ &-\frac{\alpha_k(2-\alpha_k)}{\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2 \\ &+\frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}^{k+1}-\boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1}-\boldsymbol{z}^k\right\rangle. \end{aligned} \tag{D.3}$$

Adding this inequality to (C.12), the quadratic terms combine as

$$\frac{\alpha_k(2-\alpha_k)}{2\gamma} - \frac{\alpha_k(2-\alpha_k)}{\gamma} = -\frac{\alpha_k(2-\alpha_k)}{2\gamma}.$$

The two inner products combine into

$$\frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1}-\boldsymbol{z}^k\right\rangle.$$

Substitution into (C.12) proves (C.14). ∎

Proof of Lemma C.7. The proximal optimality condition gives $\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) \in \partial h(\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}))$, and hence $\|\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})\| \le L_h$. Substituting $\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}) = (\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) - \boldsymbol{\lambda})/\rho$ into the dual update yields

$$\boldsymbol{\lambda}^{k+1} = \left[1 - \tau_k\left(\frac{1}{\rho}+\frac{1}{s_k}\right)\right]\boldsymbol{\lambda}^k + \frac{\tau_k}{\rho}\boldsymbol{q}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k) + \frac{\tau_k}{s_k}\boldsymbol{\mu}^k.$$

The coefficients are nonnegative and sum to one. Together with $\boldsymbol{\mu}^{k+1} = (1-\beta_k)\boldsymbol{\mu}^k + \beta_k\boldsymbol{\lambda}^{k+1}$, this proves $\|\boldsymbol{\lambda}^k\|, \|\boldsymbol{\mu}^k\| \le M$ by induction from the definition of M. The center update then gives $\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\| \le 2M\beta_k$.

At the saddle point, the dual first-order condition gives

$$\boldsymbol{\lambda}_k^\star(\boldsymbol{z}) = \frac{s_k}{\rho + s_k}\boldsymbol{q}\big(\boldsymbol{x}_k^\star(\boldsymbol{z}), \boldsymbol{\lambda}_k^\star(\boldsymbol{z})\big) + \frac{\rho}{\rho + s_k}\boldsymbol{\mu}^k.$$

Since both vectors in this convex combination have norm at most M, $\|\boldsymbol{\lambda}_k^\star(\boldsymbol{z})\| \le M$. ∎

Proof of Lemma C.8. Fix k and $\boldsymbol{z}$. Strong convexity gives the unique minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})$ for each $\boldsymbol{\lambda}$. The finite, smooth, $1/s_k$-strongly concave function $d_k(\cdot; \boldsymbol{z})$ tends to $-\infty$ as $\|\boldsymbol{\lambda}\| \to \infty$, so it has a unique maximizer. The minimax equality and strong convexity in $\boldsymbol{x}$ identify the unique saddle pair and give $\boldsymbol{x}_k^\star(\boldsymbol{z}) = \boldsymbol{x}_k(\boldsymbol{\lambda}_k^\star(\boldsymbol{z}); \boldsymbol{z})$.

The maximizing property at $\boldsymbol{x}_k^\star(\boldsymbol{z})$ and strong convexity at fixed $\boldsymbol{\lambda}$ imply

$$\begin{aligned} p_k(\boldsymbol{z}) - d_k(\boldsymbol{\lambda}; \boldsymbol{z}) &\ge \mathcal{P}_k(\boldsymbol{x}_k^\star(\boldsymbol{z}), \boldsymbol{\lambda}; \boldsymbol{z}) - \mathcal{P}_k(\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}), \boldsymbol{\lambda}; \boldsymbol{z}) \\ &\ge \frac{\mu_\gamma}{2}\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}) - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2. \end{aligned}$$

This proves (C.16). For the center sensitivity of $\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})$, add the strong-convexity inequalities at the minimizers for $\boldsymbol{z}$ and $\boldsymbol{z}'$. The only center-dependent term is $\|\boldsymbol{x} - \boldsymbol{z}\|^2/(2\gamma)$, so

$$\mu_\gamma\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}') - \boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})\|^2 \le \frac{1}{\gamma}\left\langle \boldsymbol{z}' - \boldsymbol{z}, \boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}') - \boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})\right\rangle.$$

For the saddle primal points, use the same argument for $\max_{\boldsymbol{\lambda}} \mathcal{P}_k(\cdot, \boldsymbol{\lambda}; \boldsymbol{z})$, whose strong-convexity modulus is also μ_γ. Cauchy–Schwarz gives (C.17) in both cases. The latter argument also gives

$$\langle \boldsymbol{z}' - \boldsymbol{z}, \boldsymbol{x}_k^\star(\boldsymbol{z}') - \boldsymbol{x}_k^\star(\boldsymbol{z})\rangle \ge \gamma\mu_\gamma\|\boldsymbol{x}_k^\star(\boldsymbol{z}') - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2.$$

Together with the Danskin formula for ∇p_k and $2\gamma\mu_\gamma > 1$, this verifies the smoothness bound used above:

$$\begin{aligned} \gamma^2\|\nabla p_k(\boldsymbol{z}') - \nabla p_k(\boldsymbol{z})\|^2 &\le \|\boldsymbol{z}' - \boldsymbol{z}\|^2 - (2\gamma\mu_\gamma - 1)\|\boldsymbol{x}_k^\star(\boldsymbol{z}') - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2 \\ &\le \|\boldsymbol{z}' - \boldsymbol{z}\|^2. \end{aligned}$$

∎

Proof of Lemma C.9. Let $u_{k+1} := (\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{z}^{k+1})$ and define

$$\widehat{\Psi}_{k+1|k} := \mathcal{P}_{k+1}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - 2d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) + 2p_k(\boldsymbol{z}^{k+1}).$$

Then

$$\Psi_{k+1}(u_{k+1}) - \Psi_k(u_k) = \widehat{\Psi}_{k+1|k} - \Psi_k(u_k) + R_k,$$

where

$$R_k := 2\big[d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - d_{k+1}(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1})\big] + 2\big[p_{k+1}(\boldsymbol{z}^{k+1}) - p_k(\boldsymbol{z}^{k+1})\big].$$

Apply Lemma C.6 to the first difference. Decompose the indefinite term at $\boldsymbol{x}_k^\star(\boldsymbol{z}^{k+1})$ and apply Lemma C.8, followed by Young's inequality. The four bounds in (C.18) absorb the resulting cross terms and leave

$$\begin{aligned} \widehat{\Psi}_{k+1|k} - \Psi_k(u_k) \le\ & -\frac{1}{4\eta_k}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{1}{8\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 - \frac{\alpha_k}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \\ &+ \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 \\ &+ \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2. \end{aligned}$$

It remains to bound R_k. Lemma C.7 gives $\|\boldsymbol{\lambda}^j\|, \|\boldsymbol{\mu}^j\|, \|\boldsymbol{\lambda}_j^\star(\boldsymbol{z})\| \le M$ and $\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\| \le 2M\beta_k$. Direct comparison of the two quadratic damping terms gives, uniformly in $\boldsymbol{x}$ and $\boldsymbol{z}$,

$$|d_{k+1}(\boldsymbol{\lambda}; \boldsymbol{z}) - d_k(\boldsymbol{\lambda}; \boldsymbol{z})| \le \Delta_k, \qquad |p_{k+1}(\boldsymbol{z}) - p_k(\boldsymbol{z})| \le \Delta_k,$$

where

$$\Delta_k := 2M^2 \left(\frac{1}{s_k} - \frac{1}{s_{k+1}} \right) + \frac{2M}{s_k} \|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|.$$

Consequently,

$$|R_k| \le 4\Delta_k \le 8M^2 \left(\frac{1}{s_k} - \frac{1}{s_{k+1}} \right) + 16M^2 \frac{\beta_k}{s_k}.$$

The two explicit positive terms above satisfy

$$\frac{\beta_k(2-\beta_k)}{2s_k} \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 \le 4M^2 \frac{\beta_k}{s_k},$$

and

$$\frac{1}{2} \left(\frac{1}{s_k} - \frac{1}{s_{k+1}} \right) \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2 \le 2M^2 \left(\frac{1}{s_k} - \frac{1}{s_{k+1}} \right).$$

Adding the estimates proves (C.20).

For the stated schedule, the two error series satisfy

$$\sum_{k=0}^{\infty} \left(\frac{1}{s_k} - \frac{1}{s_{k+1}} \right) = \frac{1}{c_s}, \qquad \sum_{k=0}^{\infty} \frac{\beta_k}{s_k} \le \frac{4\theta_\mu}{c_s},$$

which makes the accumulated transport error finite. The lower bound (C.19) follows as follows. For any $\boldsymbol{\lambda} \in \partial h(\mathcal{A}(\boldsymbol{x}))$, proximal optimality gives $\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}) = \mathcal{A}(\boldsymbol{x})$ and hence $\mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) = F(\boldsymbol{x})$. Since $\|\boldsymbol{\lambda}\| \le L_h$ and $\|\boldsymbol{\mu}^k\| \le M$,

$$\max_{\boldsymbol{\lambda}} \mathcal{P}_k(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}) \ge F(\boldsymbol{x}) + \frac{1}{2\gamma} \|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{(L_h + M)^2}{2s_k}.$$

Minimizing in $\boldsymbol{x}$ and using $s_k \ge s_0$ proves (C.19), because $\Psi_k = (\mathcal{P}_k - d_k) + (p_k - d_k) + p_k \ge p_k$. Summing (C.20) and using the error budget then gives

$$\begin{aligned} &\sum_{k=0}^{N-1} \left[\frac{1}{4\eta_k} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \frac{1}{8\tau_k} \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{\alpha_k}{2\gamma} \|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \right] \\ &\quad \le \Psi_0(\boldsymbol{x}^0, \boldsymbol{\lambda}^0; \boldsymbol{z}^0) - \inf_{\boldsymbol{x} \in \mathbb{R}^n} F(\boldsymbol{x}) + \frac{(L_h + M)^2}{2s_0} + \frac{(10 + 80\theta_\mu)M^2}{c_s}, \qquad N \ge 1. \end{aligned}$$

□

Proof of Theorem C.1. The schedule (C.2) satisfies (C.18): $\eta_k L_x = 1/2$, $\tau_k L_{d,k} \le \tau_k L_{d,0} \le 1/2$, $\tau_k G_{\mathcal{A}}^2 \le \mu_\gamma^2/(72L_x) = \eta_k \mu_\gamma^2/36$, and the two bounds on α_k follow from $s_k \ge s_0$ and its definition. Moreover, $s_{k+1} \ge s_k > 0$, $\eta_k, \tau_k > 0$, and $0 < \beta_k \le 1$. Since $L_{d,k} \ge 1/\rho + 1/s_k$, the multiplier-boundedness condition also holds. By assumption, $\inf_{\boldsymbol{x}} F(\boldsymbol{x}) > -\infty$. The master budget proved after Lemma C.9 is therefore finite and independent of K.

The tail has $\lfloor K/2 \rfloor \ge K/3$ elements. Since the primal and dual stepsizes are fixed and $\alpha_k \ge \alpha_{K-1} = \Theta(K^{-1/3})$ on the tail, the master budget and the definition of κ give

$$\|\Delta_{\boldsymbol{x}}^{\kappa+1}\|^2 = O(K^{-1}), \qquad \|\Delta_{\boldsymbol{\lambda}}^{\kappa+1}\|^2 = O(K^{-1}), \qquad \|\boldsymbol{x}^{\kappa+1} - \boldsymbol{z}^\kappa\|^2 = O(K^{-2/3}).$$

The output certificate follows directly from (3.2). For stationarity, the $\boldsymbol{x}$-update and $L_{\mathcal{D},\boldsymbol{x}}$-smoothness of $\mathcal{D}_\rho(\cdot, \boldsymbol{\lambda}^k)$ give

$$\|\nabla f(\boldsymbol{x}^{k+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{k+1})^\top \boldsymbol{q}^k\| \le \left(\frac{1}{\eta_k} + L_x\right) \|\Delta_{\boldsymbol{x}}^{k+1}\| + \frac{1}{\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|.$$

For feasibility, the dual update gives

$$\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1} = \frac{1}{\tau_k}\Delta_{\boldsymbol{\lambda}}^{k+1} + \frac{1}{s_k}(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k),$$

and hence

$$\|\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}\| \le \frac{1}{\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\| + \frac{2M}{s_k}.$$

Square the last two bounds using $(a+b)^2 \le 2a^2 + 2b^2$ and use $s_\kappa^{-2} = O(K^{-2/3})$. Both squared residuals are $O(K^{-2/3})$, so their norms are $O(K^{-1/3})$. All constants depend only on the problem data and fixed initialization and parameters, so $K = O(\epsilon^{-3})$ suffices. ∎

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